US012414501B2

(12) **United States Patent**
Chen et al.

(10) **Patent No.: US 12,414,501 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **SELF-MOVING LAWN MOWER AND SUPPLEMENTARY OPERATION METHOD FOR AN UNMOWED REGION THEREOF**

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(73) Assignee: **Nanjing Chevron Industry Co., Ltd.**, Nanjing (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 886 days.

(21) Appl. No.: **17/665,951**

(22) Filed: **Feb. 7, 2022**

(65) **Prior Publication Data**

US 2022/0151147 A1 May 19, 2022

Related U.S. Application Data

(63) Continuation of application No. PCT/CN2020/111632, filed on Aug. 27, 2020.

(30) **Foreign Application Priority Data**

Aug. 27, 2019 (CN) 201910793036.2
Sep. 12, 2019 (CN) 201910865102.2
Dec. 31, 2019 (CN) 201911411522.X

(51) **Int. Cl.**
A01D 34/00 (2006.01)
A01D 34/78 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***A01D 34/008*** (2013.01); ***A01D 34/78*** (2013.01); ***A01D 69/02*** (2013.01); ***G01S 19/47*** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC A01D 34/008; A01D 34/78; A01D 69/02; G01S 19/47; G05D 1/0219; G05D 1/0221; G05D 1/0274
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

12,038,752 B1 * 7/2024 Blanton G05D 1/0088
2018/0004217 A1 1/2018 Biber et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 103839069 A 6/2014
CN 103891463 A 7/2014
(Continued)

OTHER PUBLICATIONS

ISA/CN, International Search Report issued on PCT application No. PCT/CN2020/111632, dated Dec. 1, 2020, 5 pages.
(Continued)

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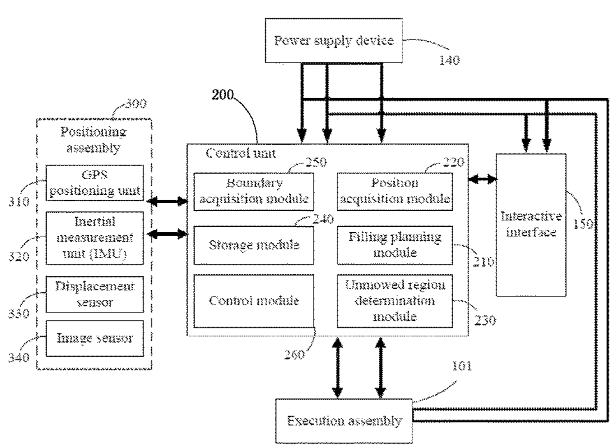
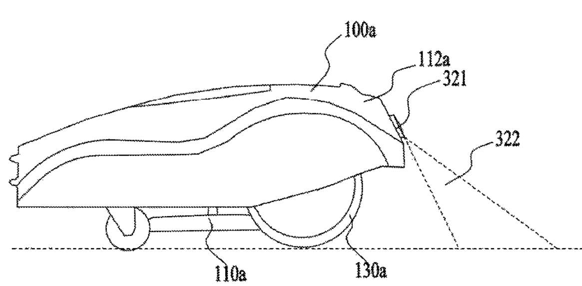
Assistant Examiner — Robert E Pezzuto
(74) *Attorney, Agent, or Firm* — Greenberg Traurig, LLP

(57) **ABSTRACT**

A self-moving mowing system includes a self-moving lawn mower and a control unit. The control unit includes a boundary acquisition module, an unmowed region determination module, a filling planning module, and a control module. The boundary acquisition module is configured to acquire information about an operation boundary so as to control the self-moving lawn mower to operate within the operation boundary. The unmowed region determination module is configured to identify information about unmowed regions within the operation boundary. The filling planning module is configured to generate an operation route along which mowing is sequentially performed in at least one unmowed region among all the unmowed regions. The
(Continued)





control module is configured to control the self-moving lawn mower to mow in the at least one unmowed region among all the unmowed regions according to the operation route.

15 Claims, 28 Drawing Sheets

(51) **Int. Cl.**

A01D 69/02	(2006.01)
G01D 1/00	(2006.01)
G01D 1/02	(2006.01)
G01S 19/47	(2010.01)
G05D 1/00	(2006.01)
A01D 101/00	(2006.01)

(52) **U.S. Cl.**
CPC ***G05D 1/0219*** (2013.01); *A01D 2101/00* (2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2020/0050208	A1*	2/2020	Frick	G05D 1/246
2021/0096574	A1*	4/2021	Lee	G05D 1/0225

FOREIGN PATENT DOCUMENTS

CN	105988471	A	10/2016
CN	106155053	A	11/2016
CN	107390698	A	11/2017
CN	107637255	A	1/2018
CN	107831773	A	3/2018
CN	107966725	A	4/2018
CN	108490932	A	9/2018
CN	108781704	A	11/2018
CN	109634287	A	4/2019
CN	110168465	A	8/2019
GB	2285905	A	8/1995
IT	201600116383	A	5/2018
WO	2014027945	A1	2/2014
WO	2018/014838	A1	1/2018
WO	2018001358	A1	1/2018

OTHER PUBLICATIONS

ISA/CN, English translation of International Search Report issued on PCT application No. PCT/CN2020/111632, dated Dec. 1, 2020, 3 pages.
ISA/CN, Written Opinion issued on PCT application No. PCT/CN2020/111632, dated Dec. 1, 2020, 3 pages.
ISA/CN, English translation of Written Opinion issued on PCT application No. PCT/CN2020/111632, dated Dec. 1, 2020, 3 pages.
Liu, Rong et al., Path Planning for UAV Based on Improved Chaotic Ant Colony Algorithm (CACA), Command Information System and Technology), vol. 9, No. 6, Jan. 14, 2019 (Jan. 14, 2019), https://portal.issn.org/resource/ISSN/1674-909X, p. 43 right-hand column and p. 44 left-hand column.
EPO, extended European search report issued on European patent application No. 20856753.7, dated Feb. 10, 2023, 8 pages.

* cited by examiner

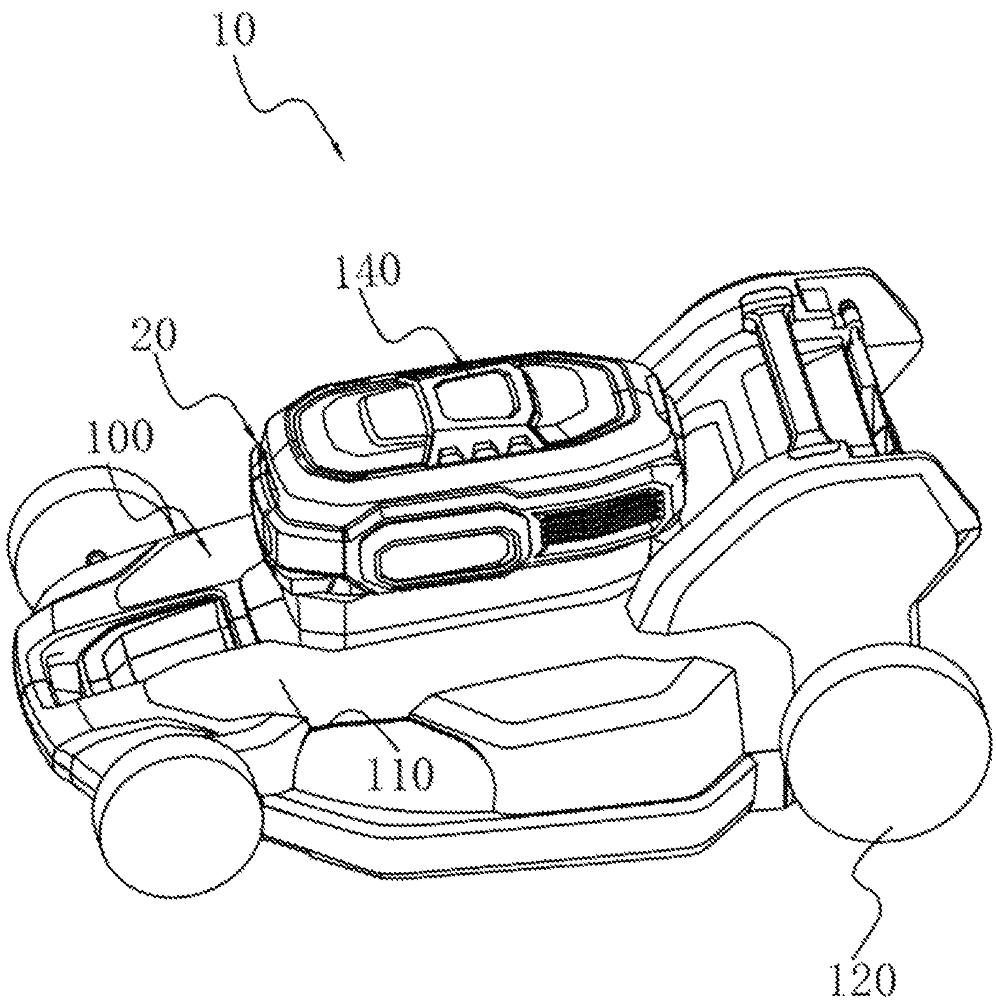


FIG. 1

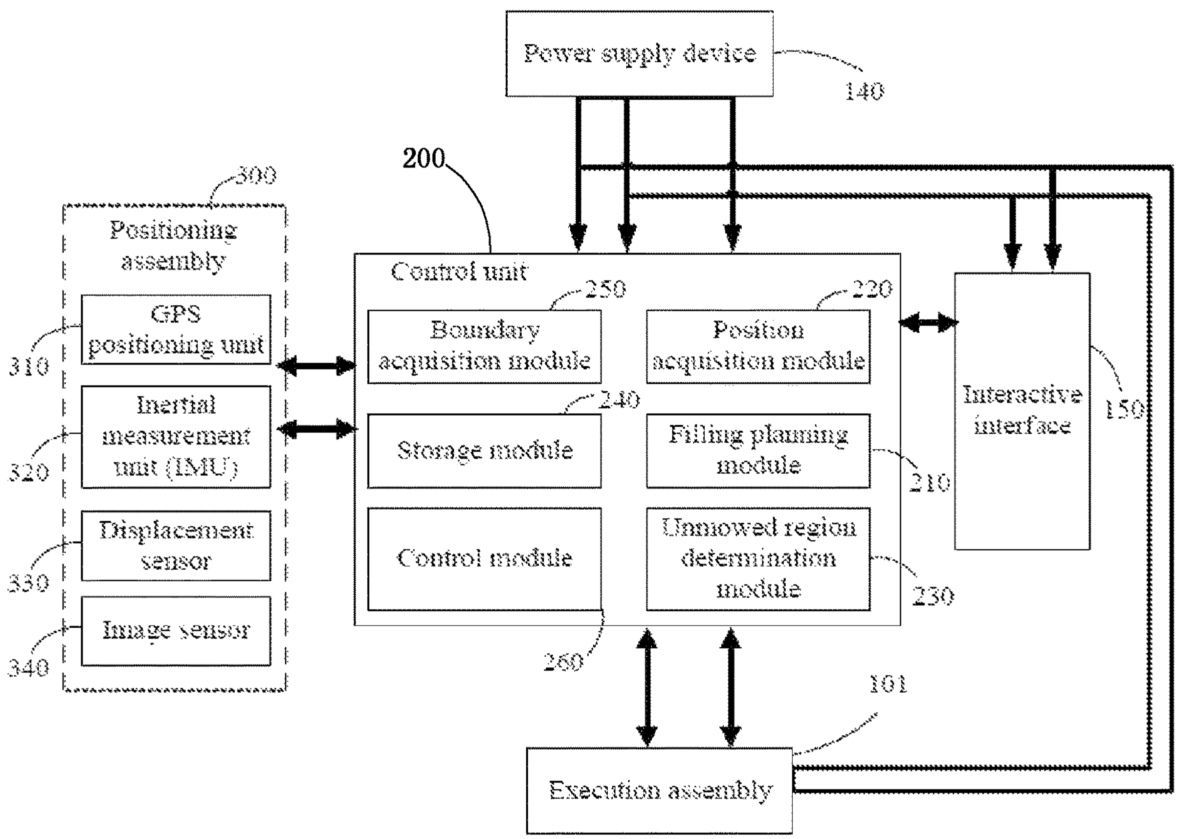

Power supply device
140
200
300
Positioning assembly
GPS positioning unit
310
Inertial measurement unit (IMU)
320
Displacement sensor
330
Image sensor
340
Control unit
250
Boundary acquisition module
220
Position acquisition module
240
Storage module
Filling planning module
210
Control module
Unmowed region determination module
230
260
Interactive interface
150
101
Execution assembly

FIG. 2A

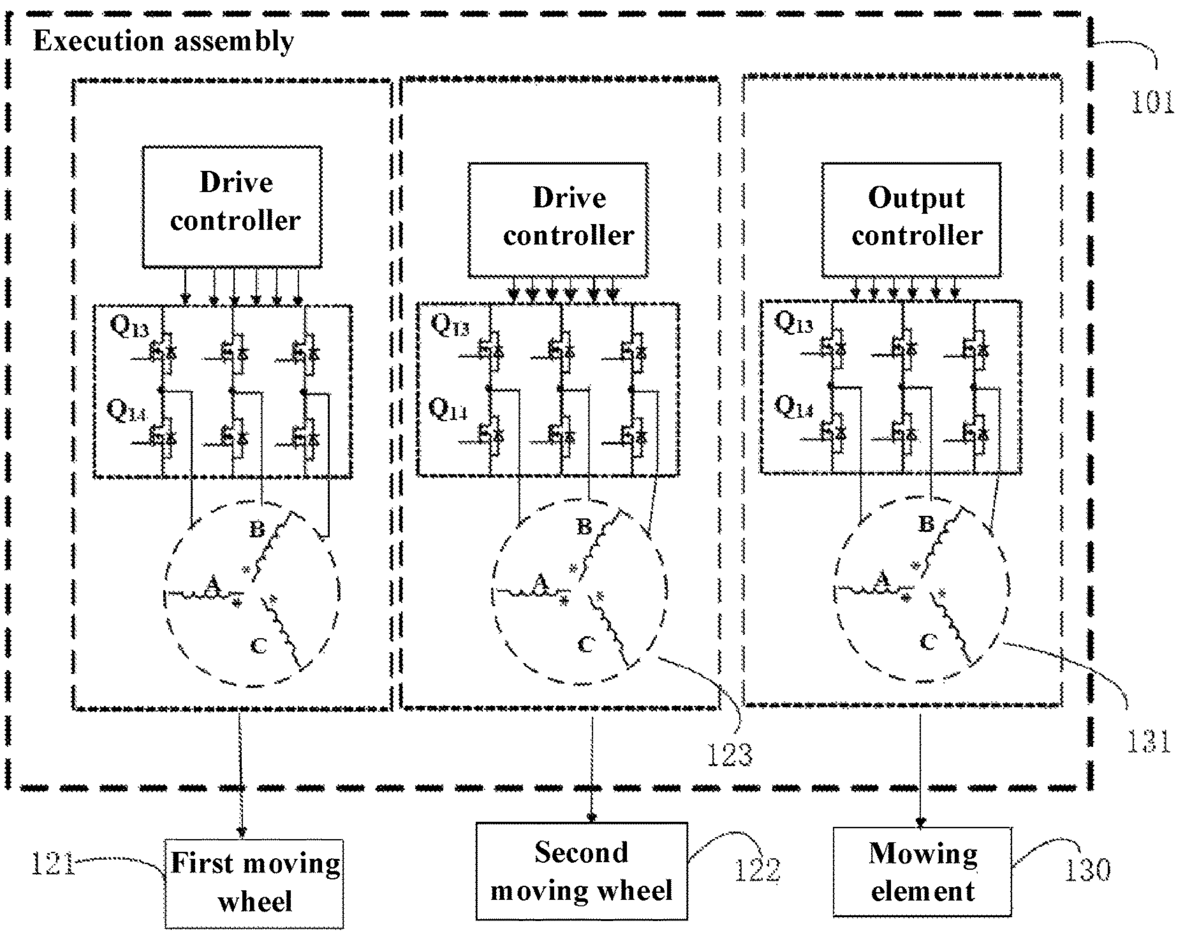

Execution assembly
Drive controller
Drive controller
Output controller
Q13
Q14
Q13
Q14
Q13
Q14
A
B
C
A
B
C
A
B
C
101
123
131
121
First moving wheel
Second moving wheel
122
Mowing element
130

FIG. 2B

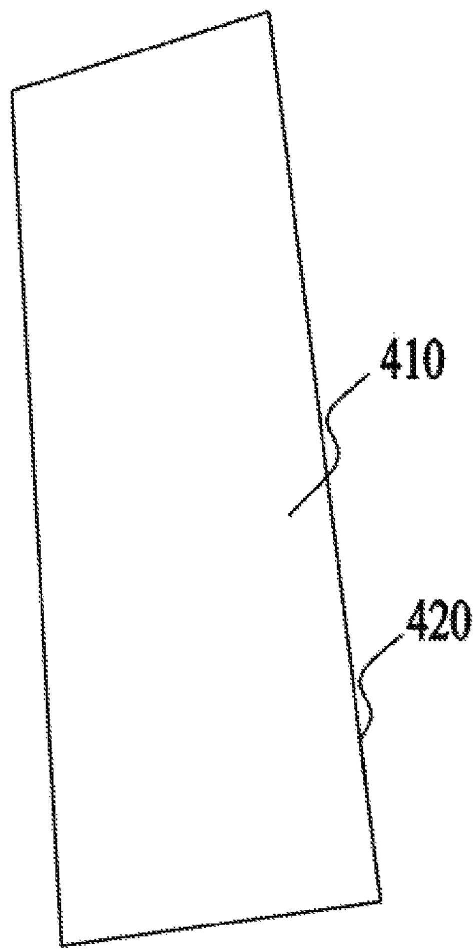


FIG. 3

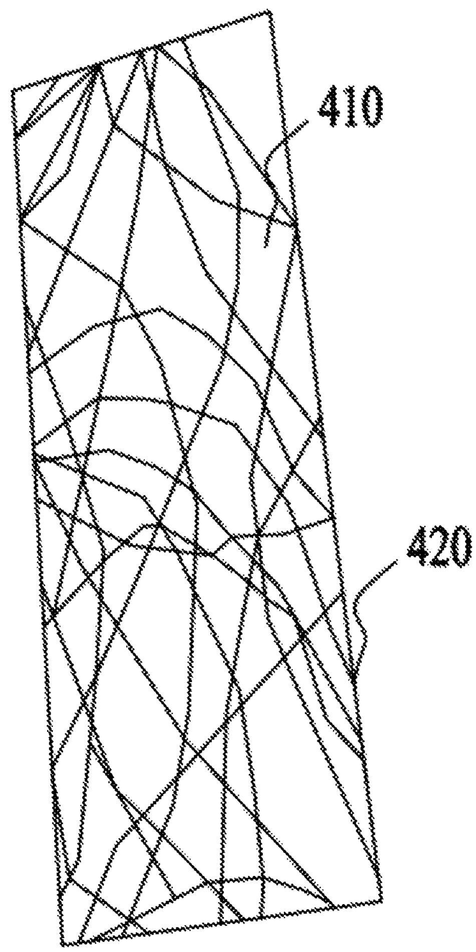


FIG. 4

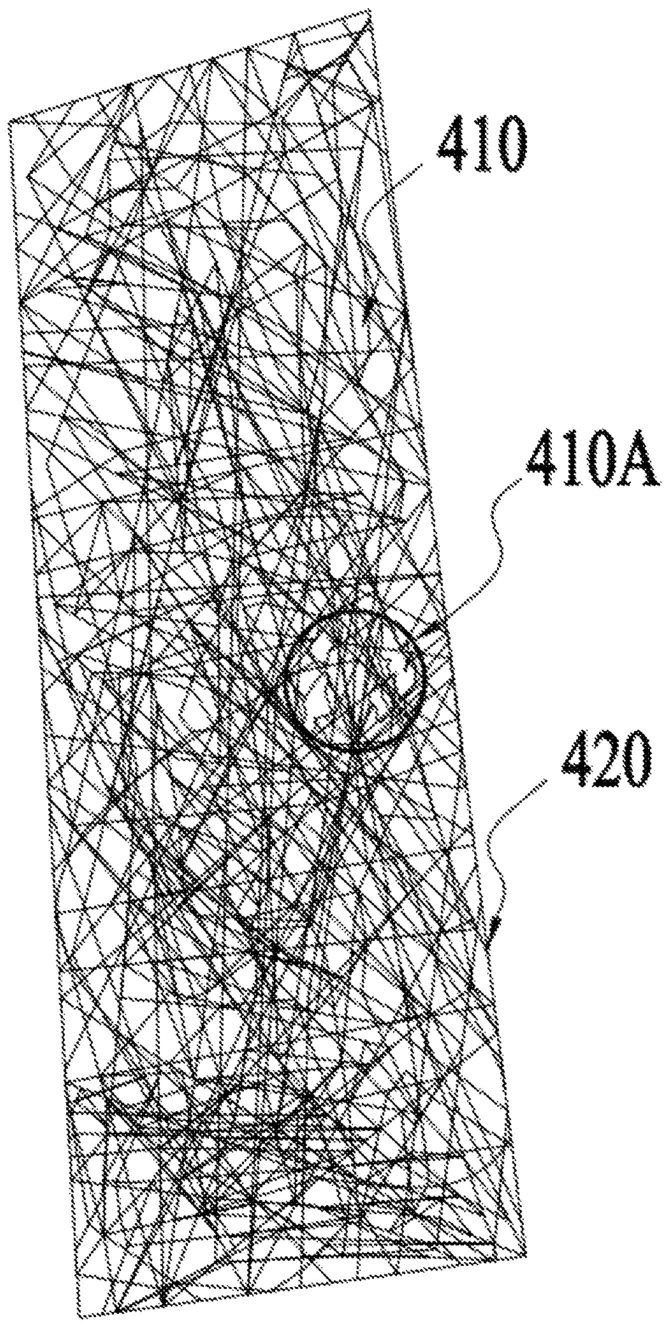


FIG. 5

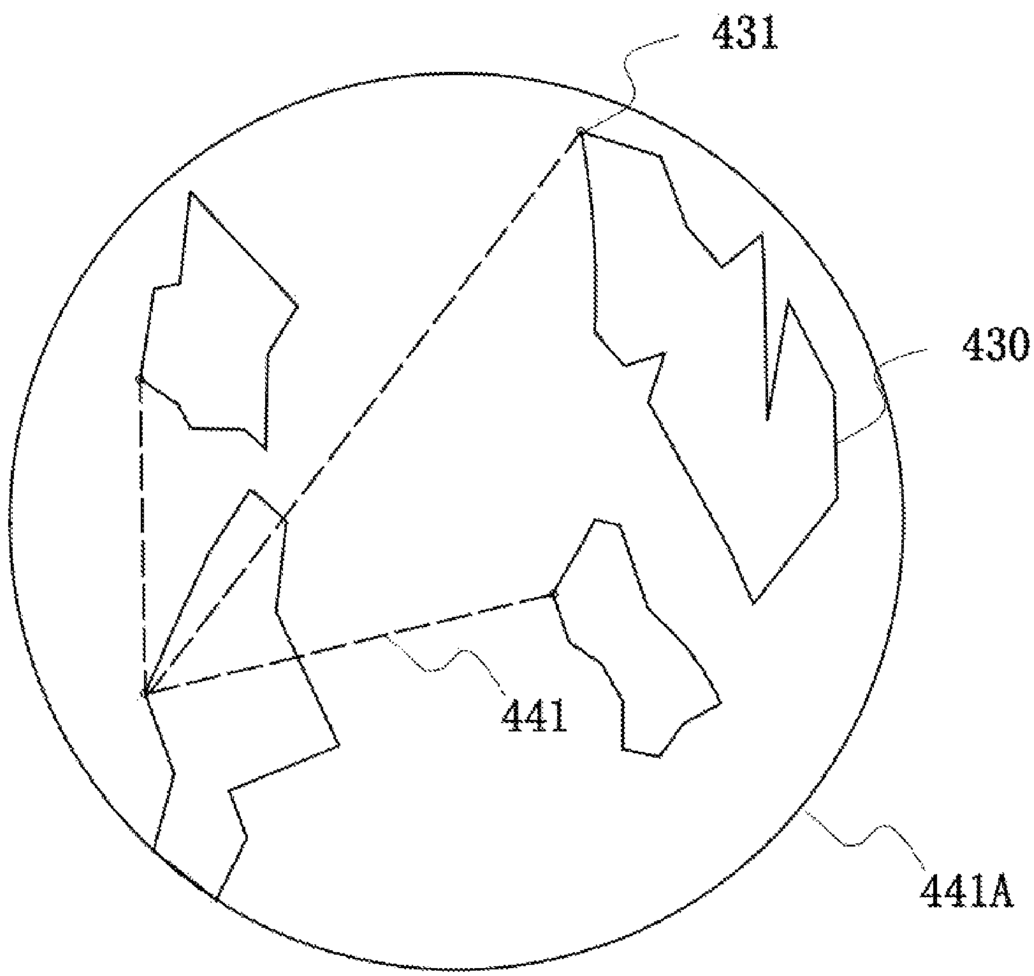


FIG. 6

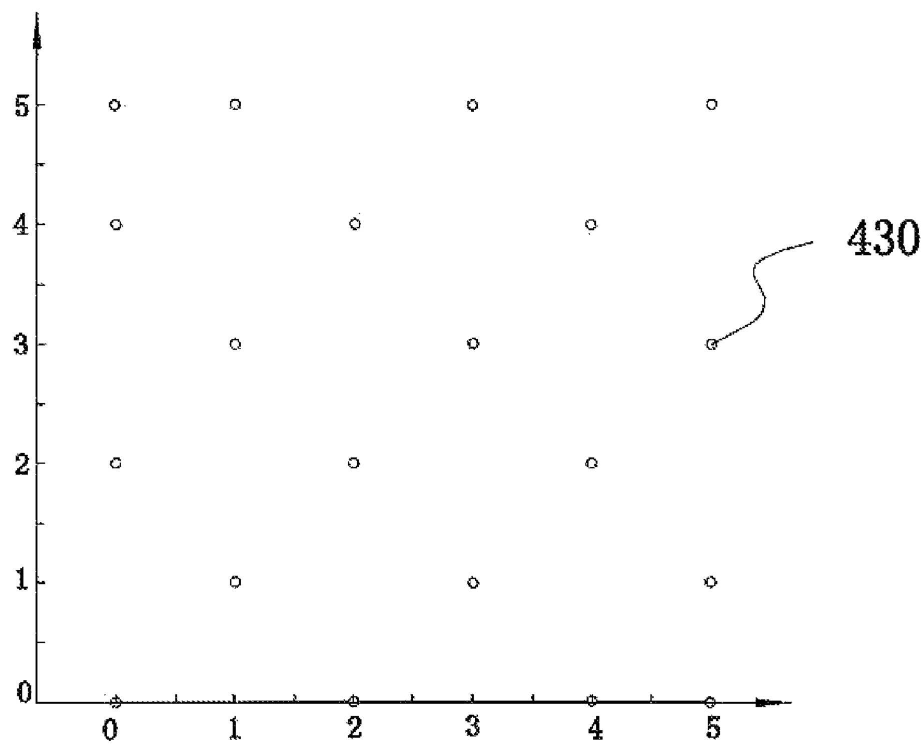


FIG. 7

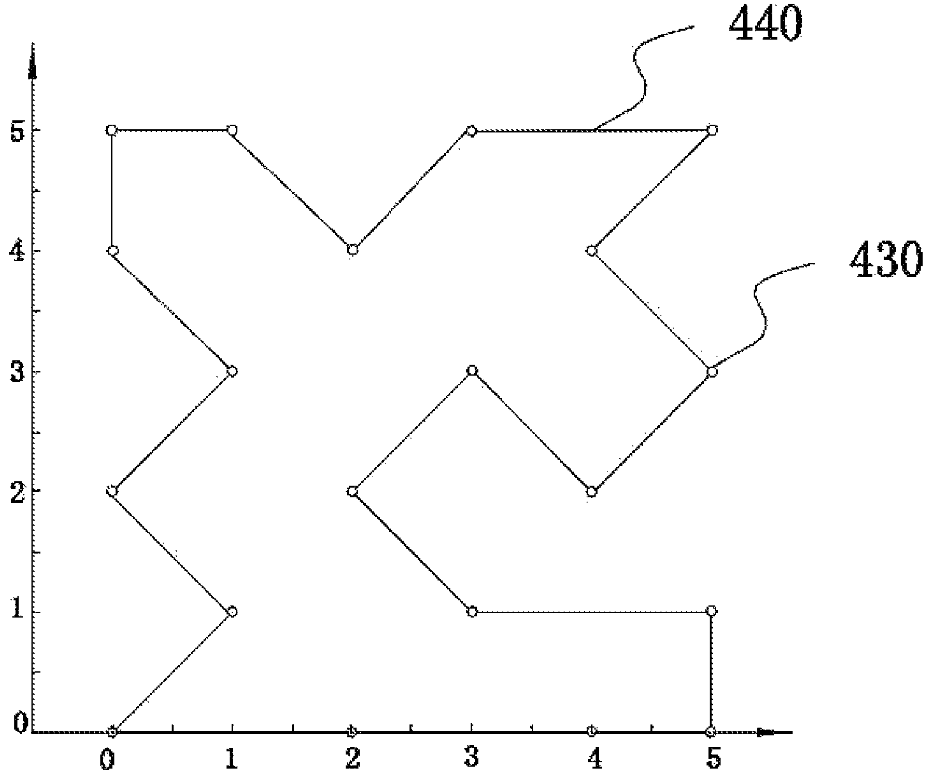
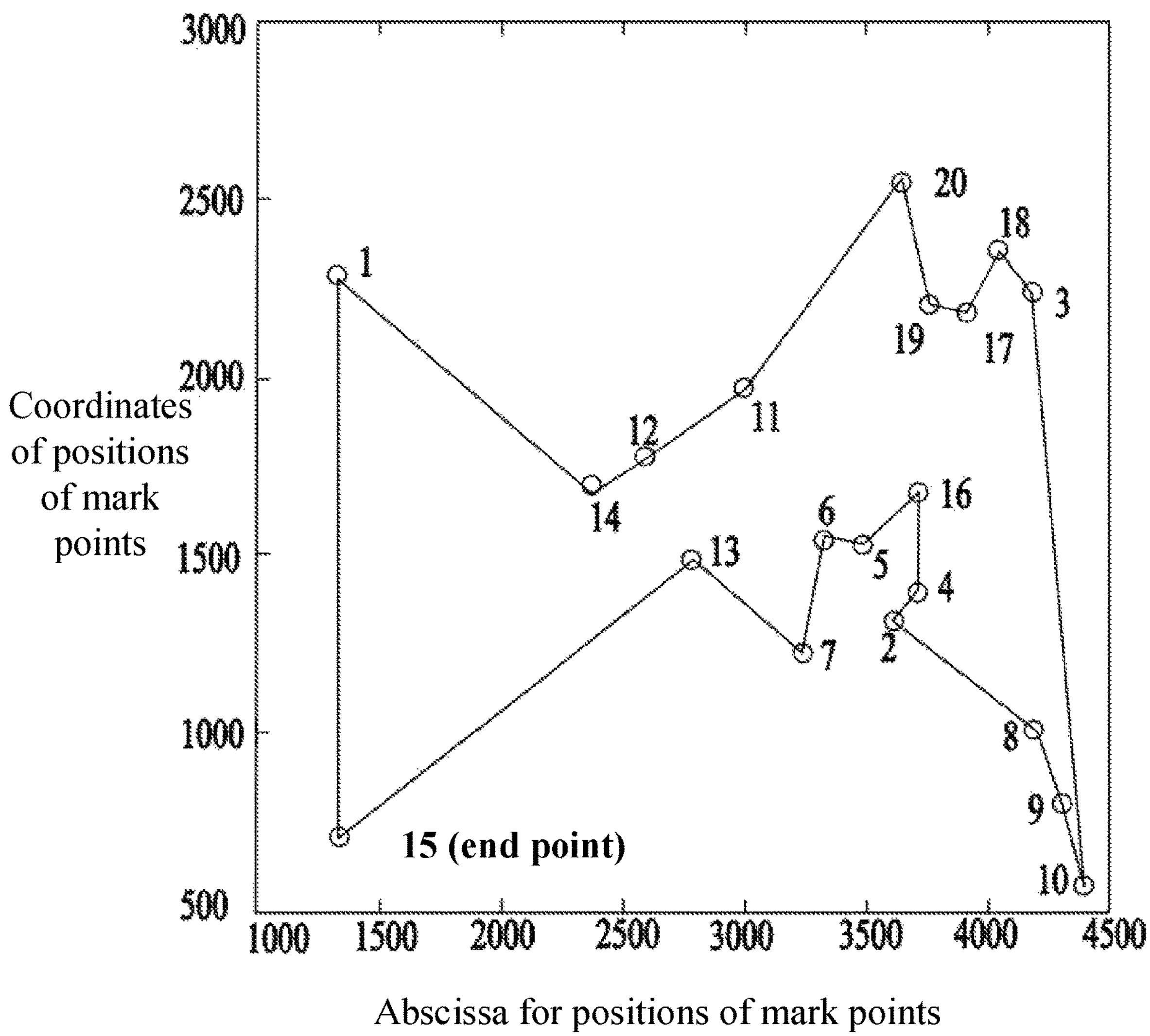


FIG. 8



3000
2500
2000
1500
1000
500
1000
1500
2000
2500
3000
3500
4000
4500
Coordinates of positions of mark points
Abscissa for positions of mark points
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15 (end point)
16
17
18
19
20

FIG. 9

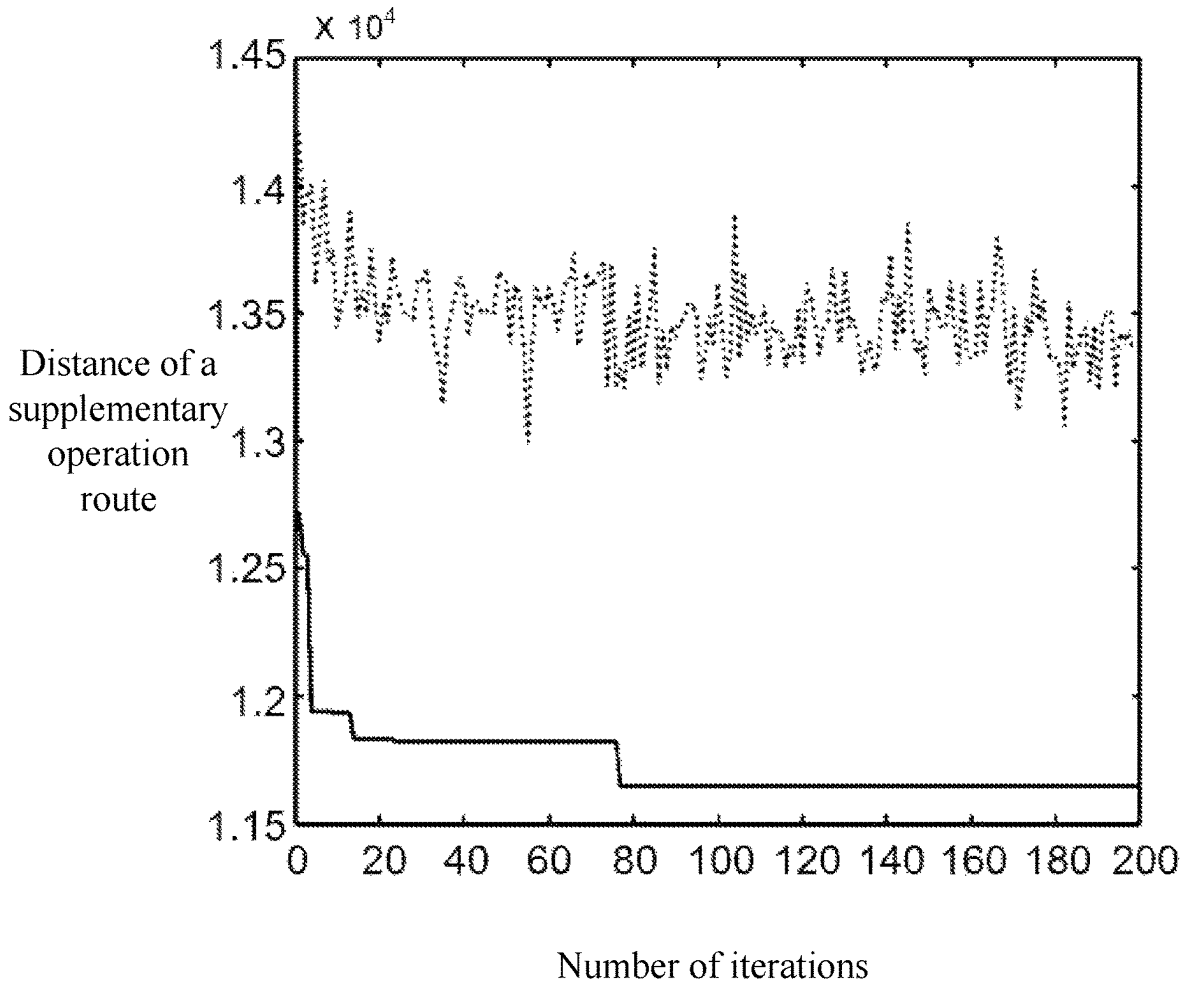

x 10^4
1.45
1.4
1.35
1.3
1.25
1.2
1.15
Distance of a supplementary operation route
0
20
40
60
80
100
120
140
160
180
200
Number of iterations

FIG. 10

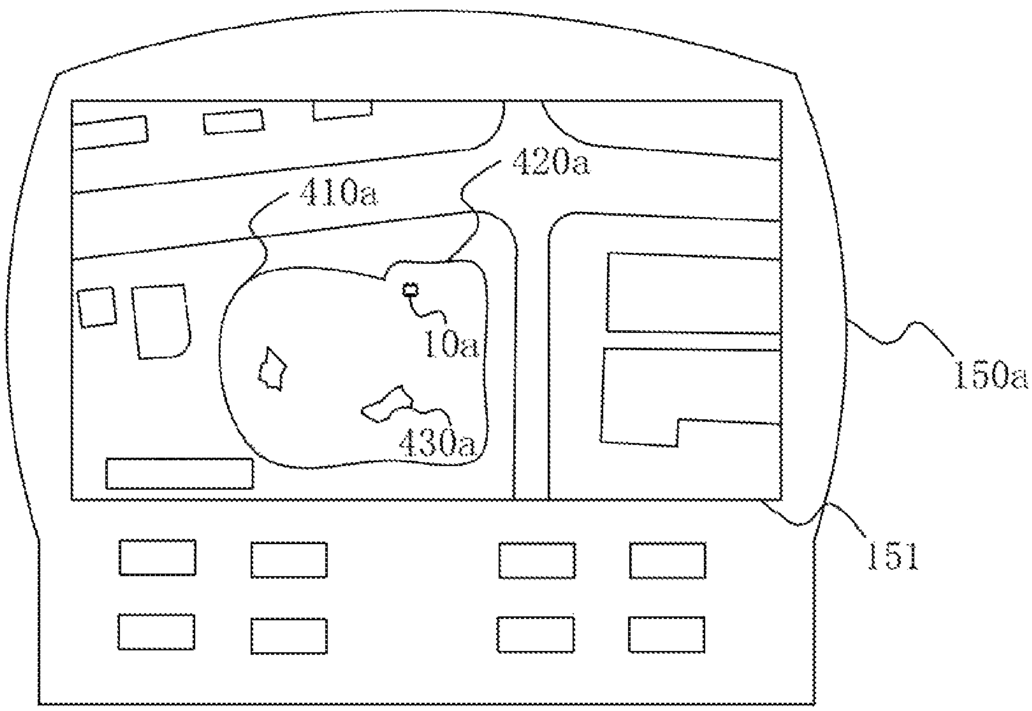


FIG. 11

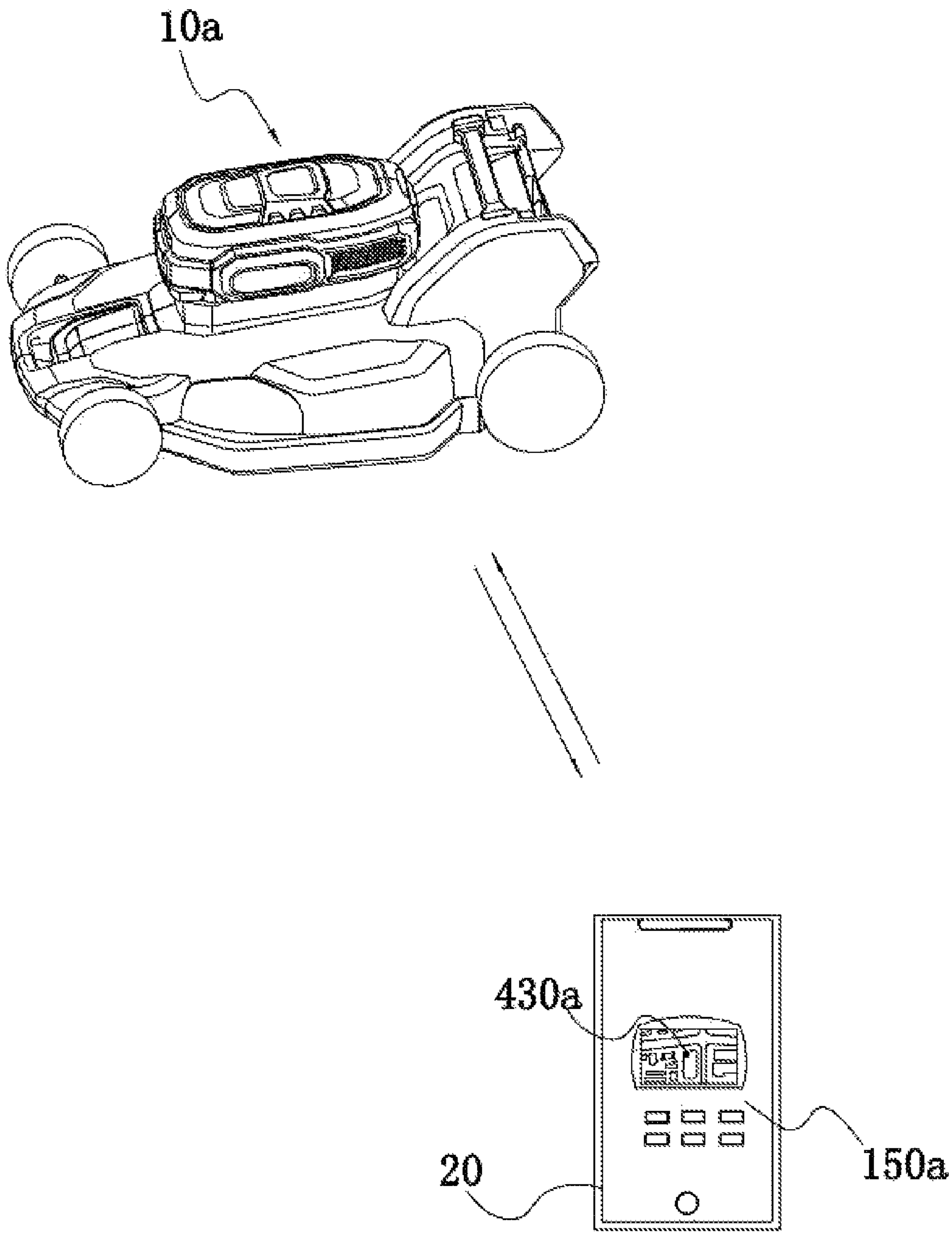


FIG. 12

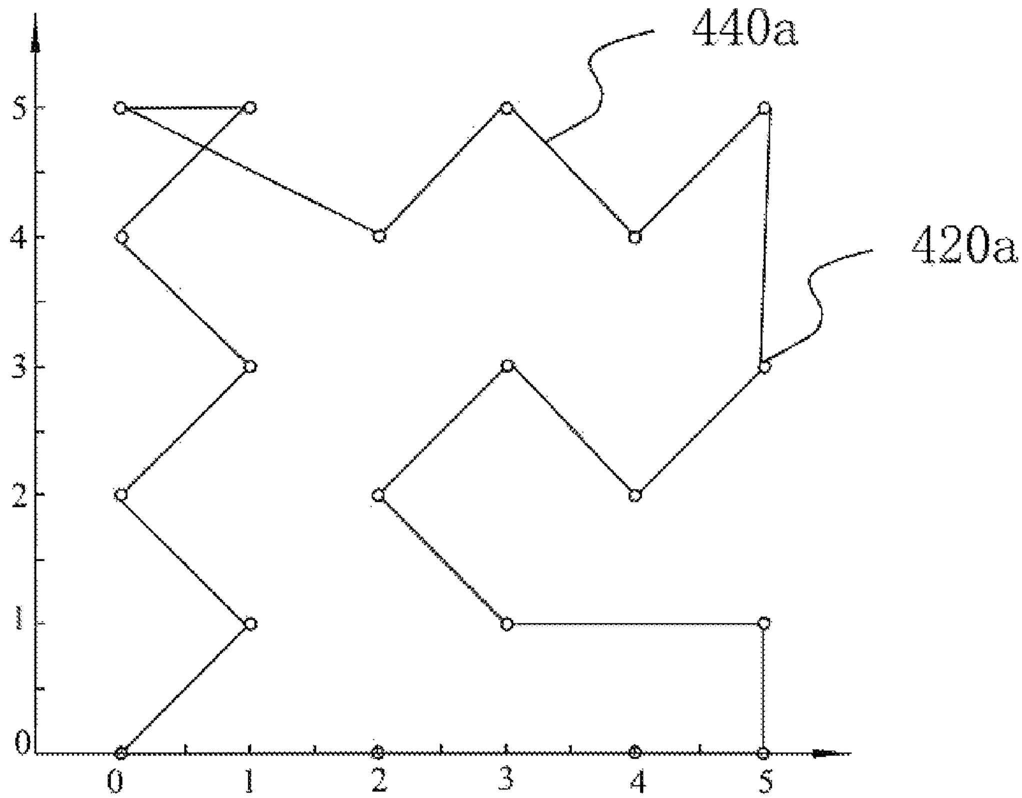


FIG. 13

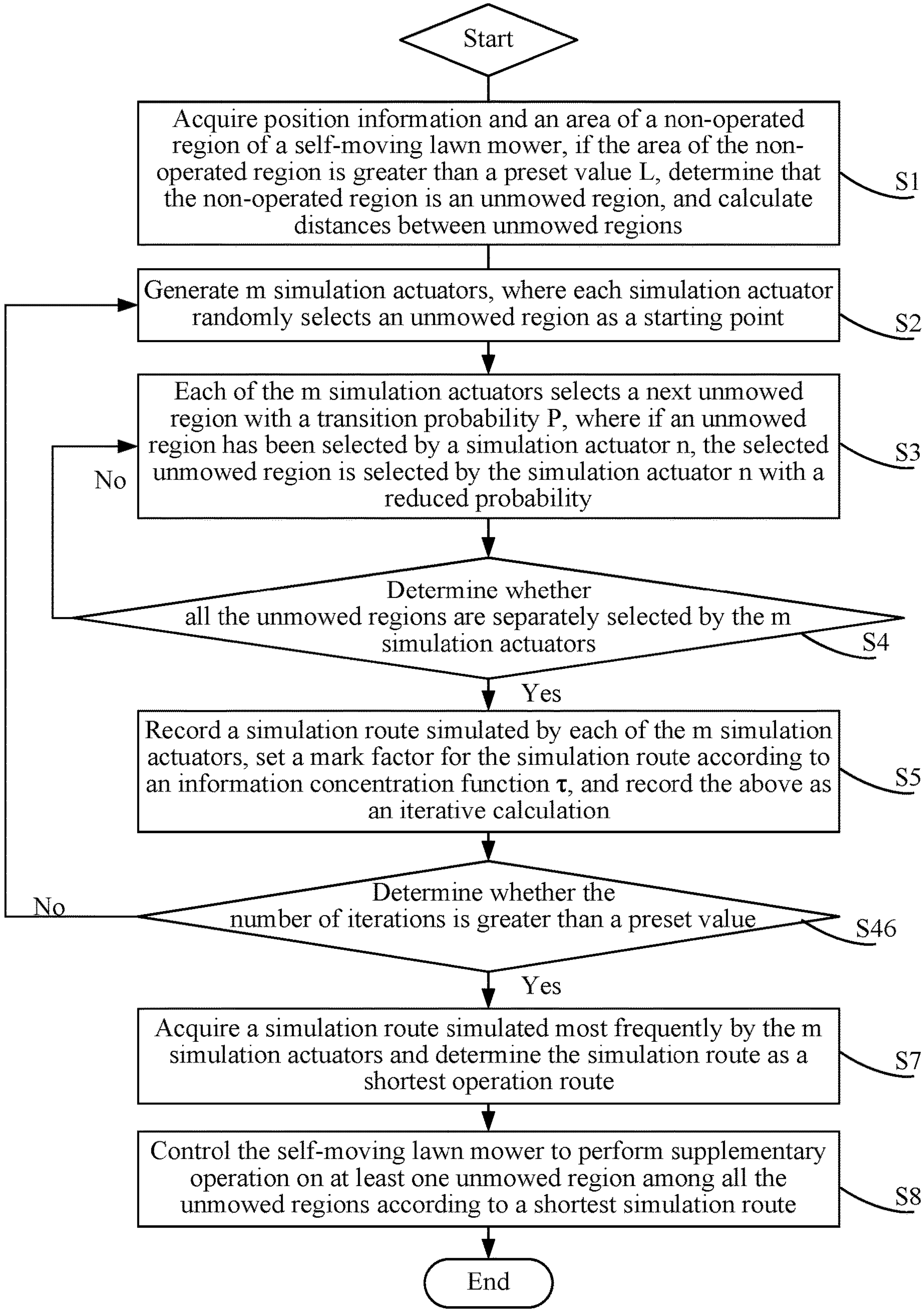

Start
Acquire position information and an area of a non-operated region of a self-moving lawn mower, if the area of the non-operated region is greater than a preset value L, determine that the non-operated region is an unmowed region, and calculate distances between unmowed regions
S1
Generate m simulation actuators, where each simulation actuator randomly selects an unmowed region as a starting point
S2
Each of the m simulation actuators selects a next unmowed region with a transition probability P, where if an unmowed region has been selected by a simulation actuator n, the selected unmowed region is selected by the simulation actuator n with a reduced probability
S3
No
Determine whether all the unmowed regions are separately selected by the m simulation actuators
S4
Yes
Record a simulation route simulated by each of the m simulation actuators, set a mark factor for the simulation route according to an information concentration function τ, and record the above as an iterative calculation
S5
No
Determine whether the number of iterations is greater than a preset value
S46
Yes
Acquire a simulation route simulated most frequently by the m simulation actuators and determine the simulation route as a shortest operation route
S7
Control the self-moving lawn mower to perform supplementary operation on at least one unmowed region among all the unmowed regions according to a shortest simulation route
S8
End

FIG. 14

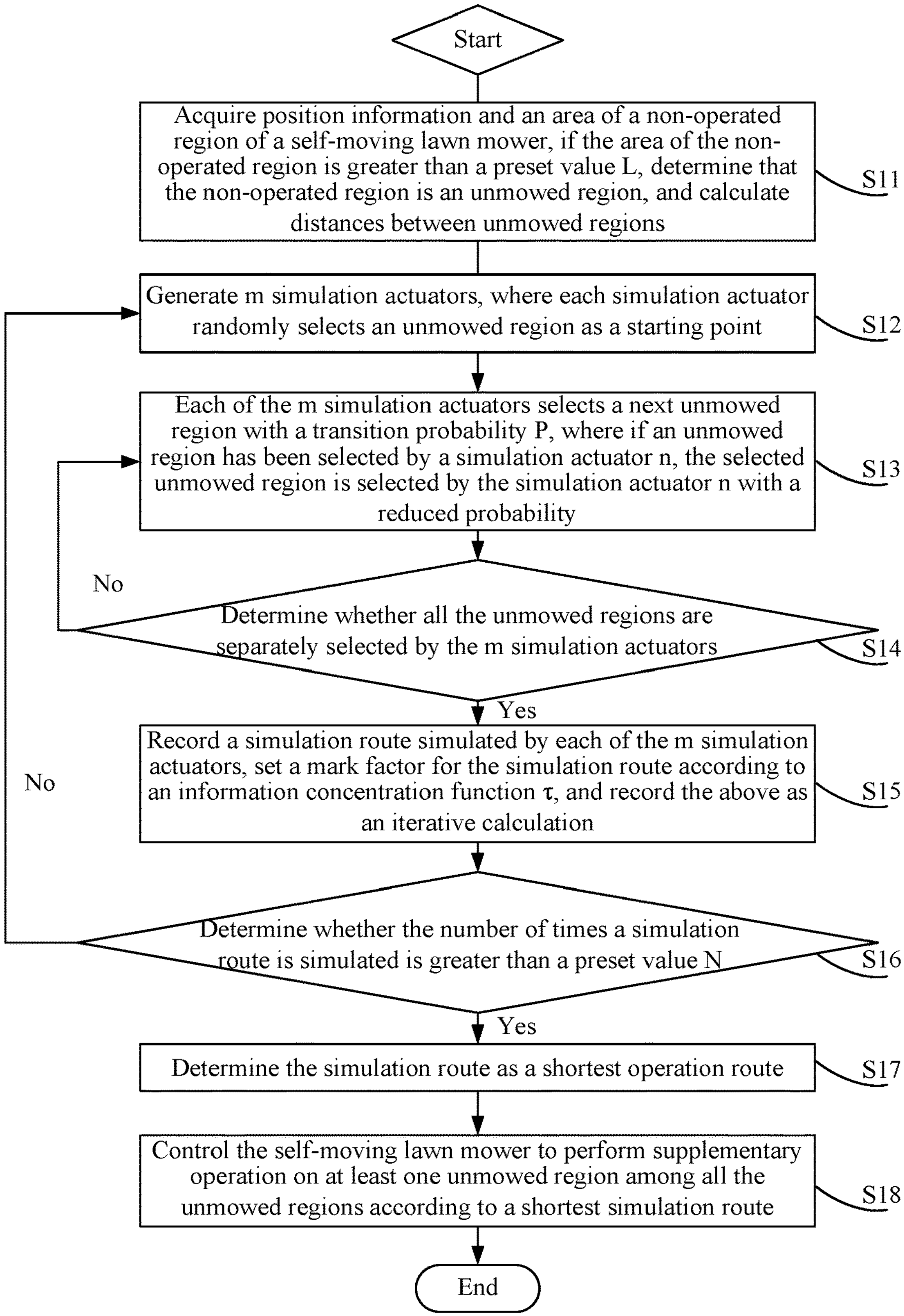

Start
Acquire position information and an area of a non-operated region of a self-moving lawn mower, if the area of the non-operated region is greater than a preset value L, determine that the non-operated region is an unmowed region, and calculate distances between unmowed regions
S11
Generate m simulation actuators, where each simulation actuator randomly selects an unmowed region as a starting point
S12
Each of the m simulation actuators selects a next unmowed region with a transition probability P, where if an unmowed region has been selected by a simulation actuator n, the selected unmowed region is selected by the simulation actuator n with a reduced probability
S13
No
Determine whether all the unmowed regions are separately selected by the m simulation actuators
S14
Yes
Record a simulation route simulated by each of the m simulation actuators, set a mark factor for the simulation route according to an information concentration function τ, and record the above as an iterative calculation
S15
No
Determine whether the number of times a simulation route is simulated is greater than a preset value N
S16
Yes
Determine the simulation route as a shortest operation route
S17
Control the self-moving lawn mower to perform supplementary operation on at least one unmowed region among all the unmowed regions according to a shortest simulation route
S18
End

FIG. 15

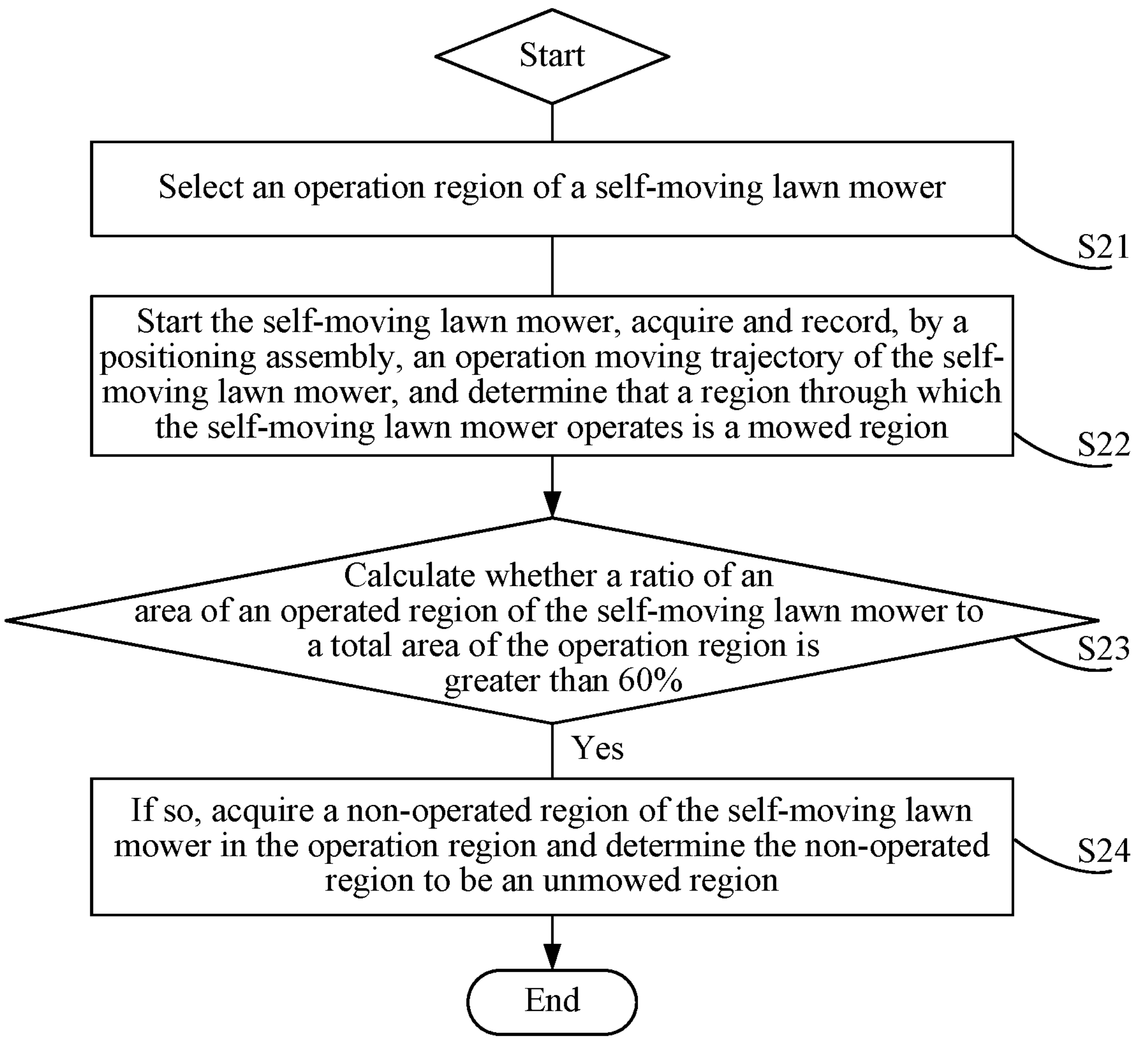
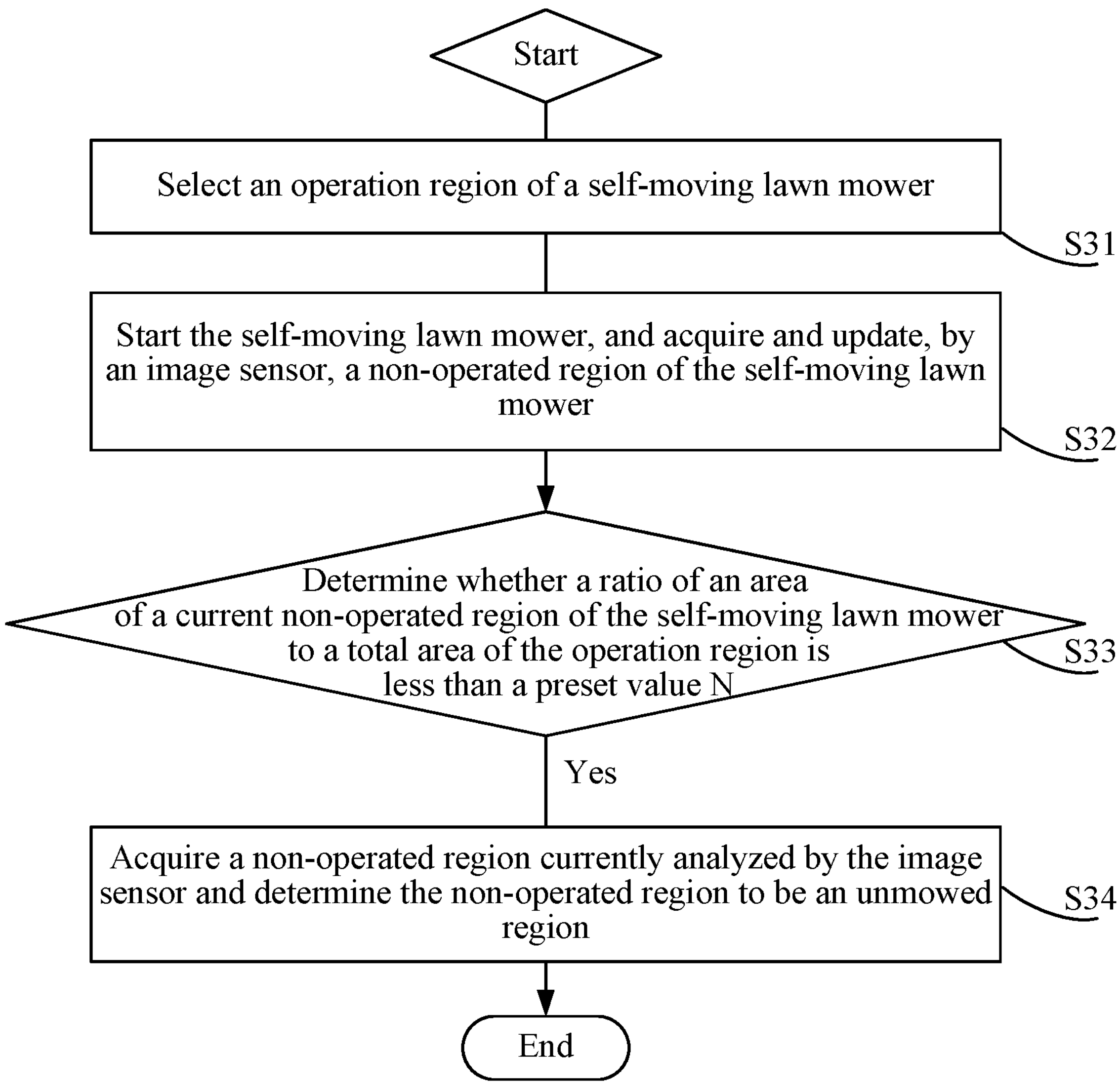

Start
Select an operation region of a self-moving lawn mower
S21
Start the self-moving lawn mower, acquire and record, by a positioning assembly, an operation moving trajectory of the self-moving lawn mower, and determine that a region through which the self-moving lawn mower operates is a mowed region
S22
Calculate whether a ratio of an area of an operated region of the self-moving lawn mower to a total area of the operation region is greater than 60%
S23
Yes
If so, acquire a non-operated region of the self-moving lawn mower in the operation region and determine the non-operated region to be an unmowed region
S24
End

FIG. 16



Start
Select an operation region of a self-moving lawn mower
S31
Start the self-moving lawn mower, and acquire and update, by an image sensor, a non-operated region of the self-moving lawn mower
S32
Determine whether a ratio of an area of a current non-operated region of the self-moving lawn mower to a total area of the operation region is less than a preset value N
S33
Yes
Acquire a non-operated region currently analyzed by the image sensor and determine the non-operated region to be an unmowed region
S34
End

FIG. 17

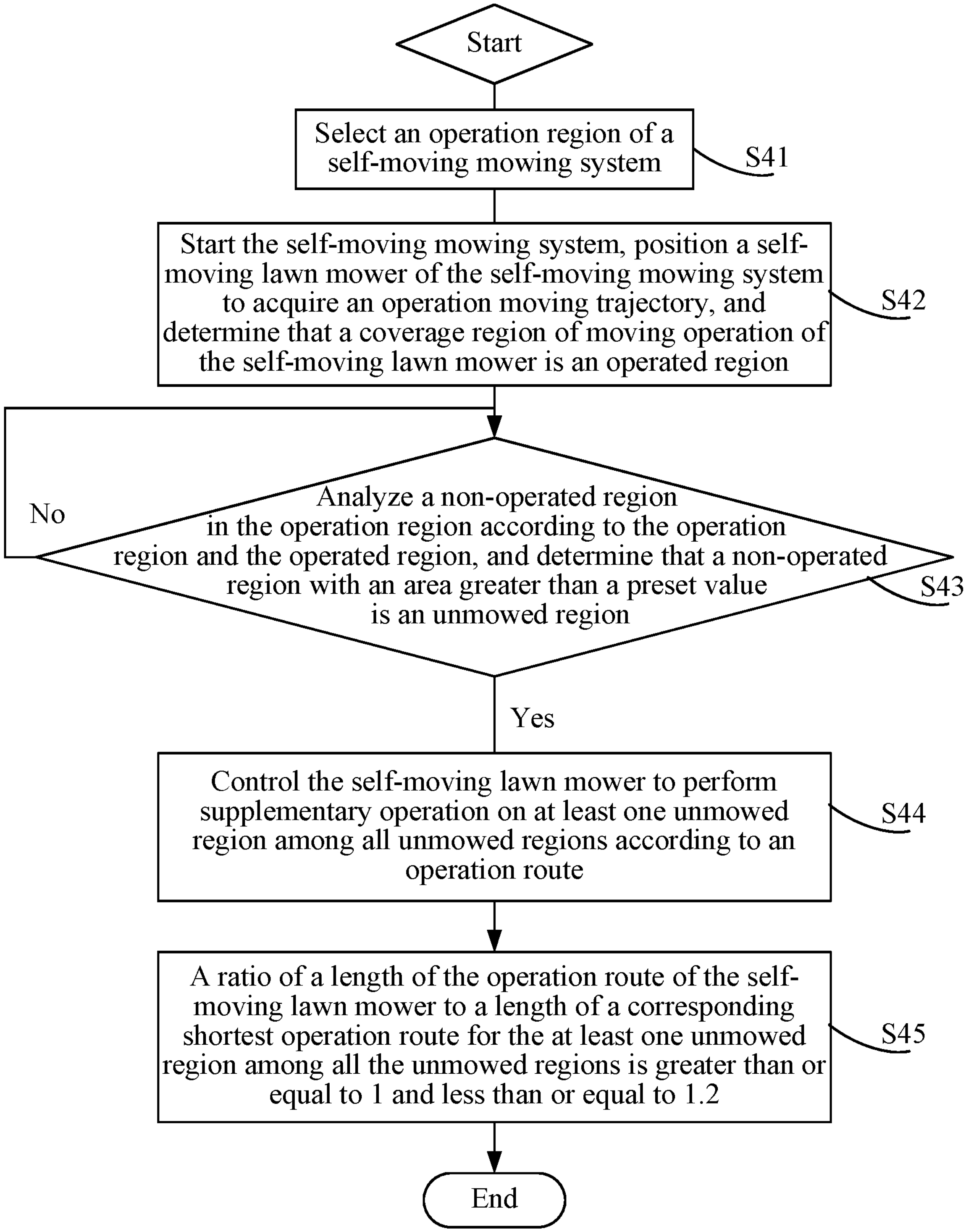

Start
Select an operation region of a self-moving mowing system
S41
Start the self-moving mowing system, position a self-moving lawn mower of the self-moving mowing system to acquire an operation moving trajectory, and determine that a coverage region of moving operation of the self-moving lawn mower is an operated region
S42
No
Analyze a non-operated region in the operation region according to the operation region and the operated region, and determine that a non-operated region with an area greater than a preset value is an unmowed region
S43
Yes
Control the self-moving lawn mower to perform supplementary operation on at least one unmowed region among all unmowed regions according to an operation route
S44
A ratio of a length of the operation route of the self-moving lawn mower to a length of a corresponding shortest operation route for the at least one unmowed region among all the unmowed regions is greater than or equal to 1 and less than or equal to 1.2
S45
End

FIG. 18

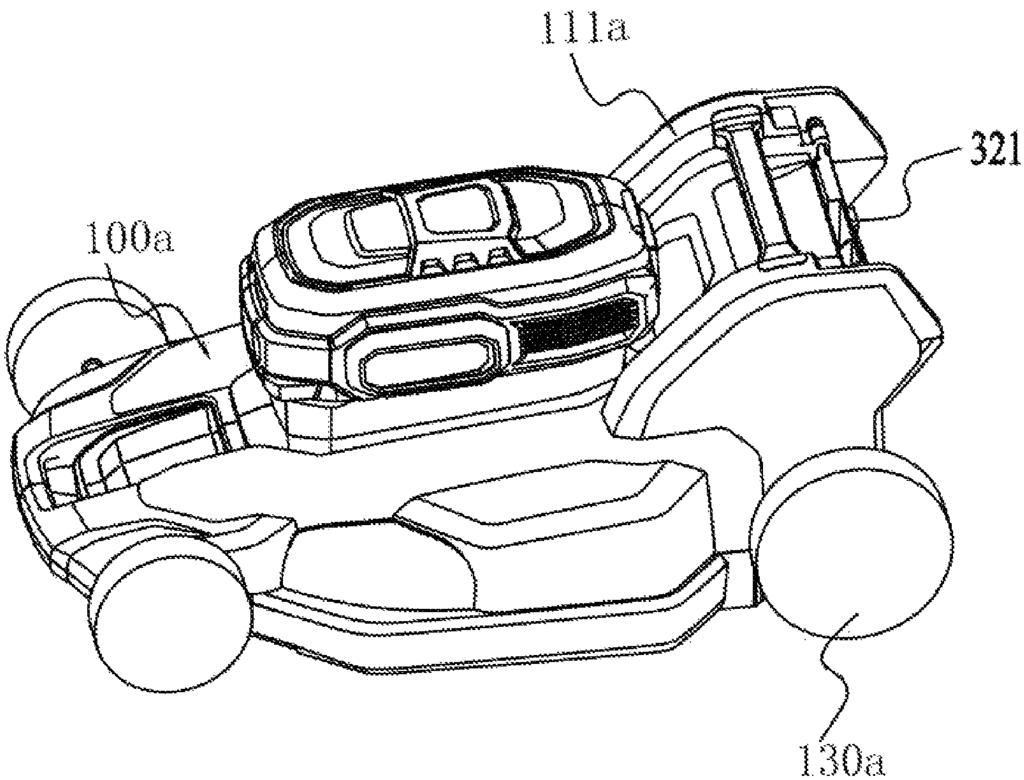


FIG. 19

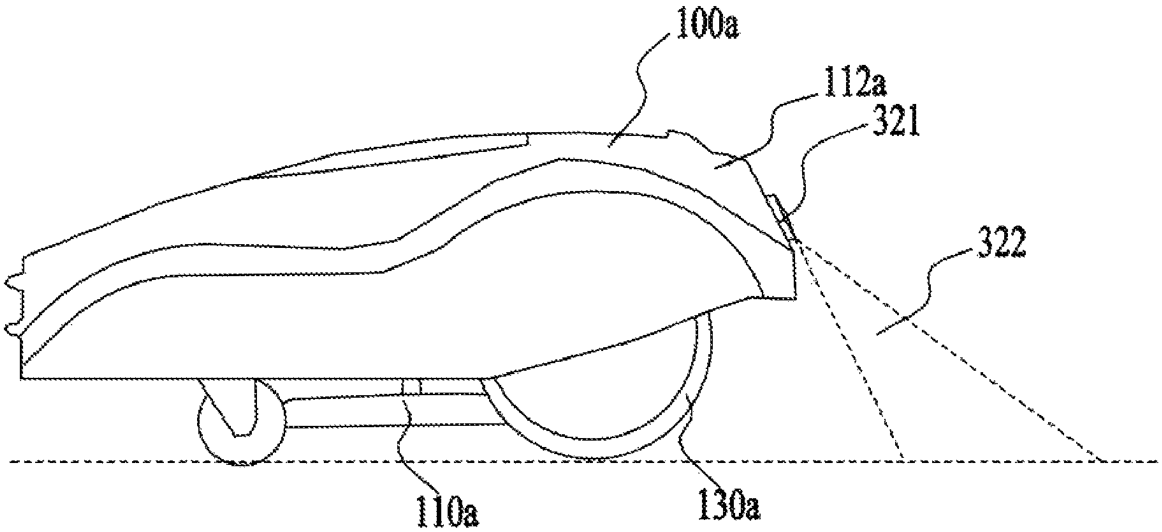


FIG. 20

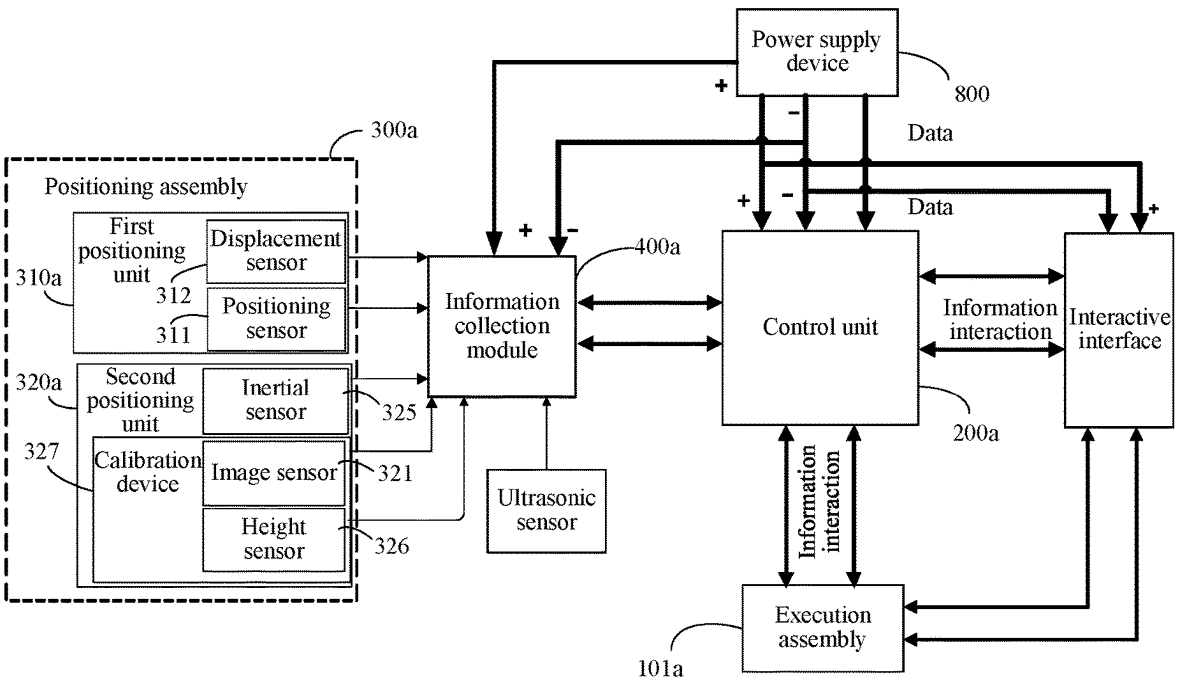

Power supply device
800
Data
Data
300a
Positioning assembly
First positioning unit
Displacement sensor
312
Positioning sensor
311
310a
320a
Second positioning unit
Inertial sensor
325
327
Calibration device
Image sensor
321
Height sensor
326
Information collection module
400a
Ultrasonic sensor
Control unit
200a
Information interaction
Interactive interface
Information interaction
Execution assembly
101a

FIG. 21A

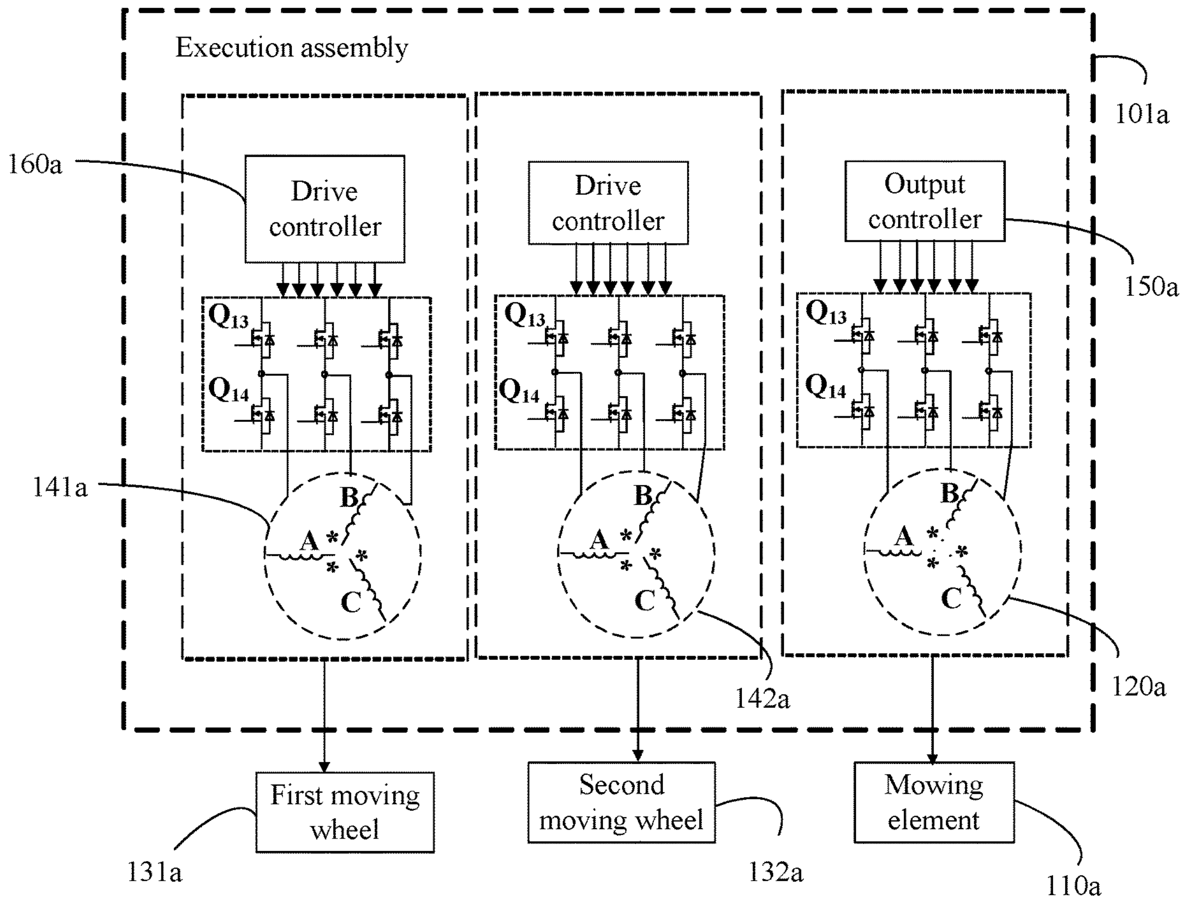

Execution assembly
101a
160a
Drive controller
Drive controller
Output controller
150a
Q13
Q14
Q13
Q14
Q13
Q14
141a
B
A
C
B
A
C
B
A
C
142a
120a
First moving wheel
Second moving wheel
Mowing element
131a
132a
110a

FIG. 21B

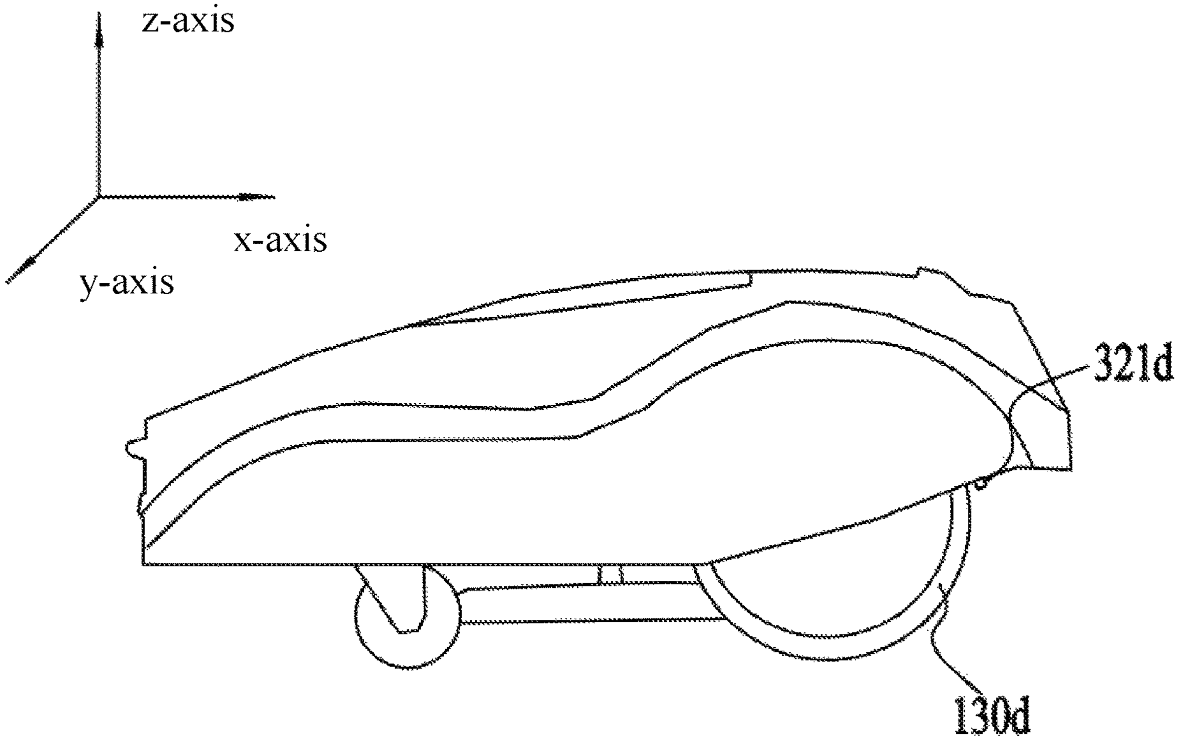


FIG. 22

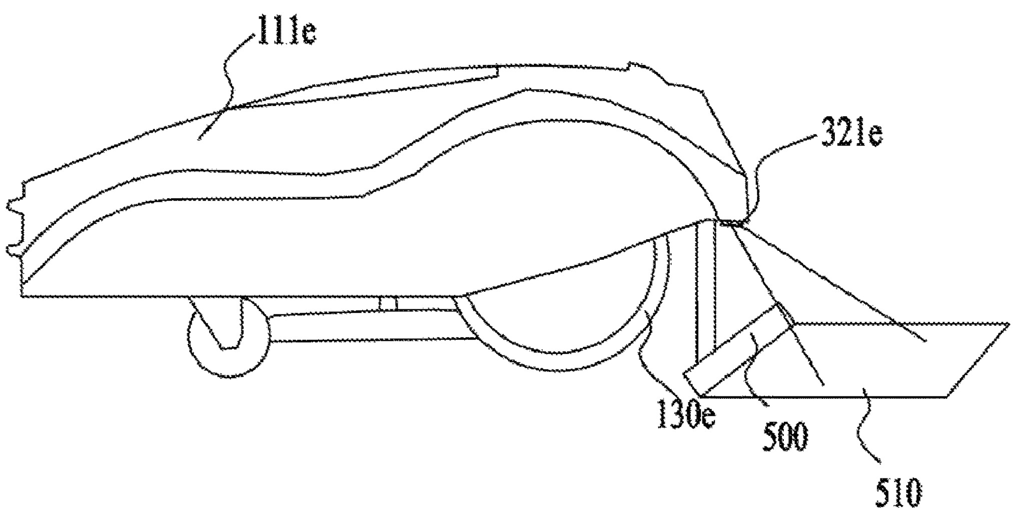


FIG. 23

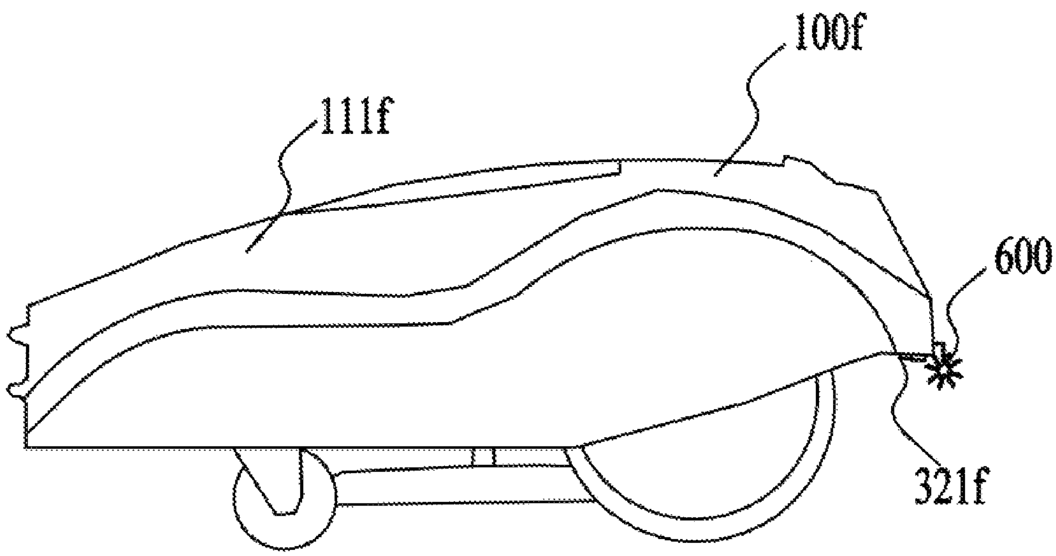


FIG. 24

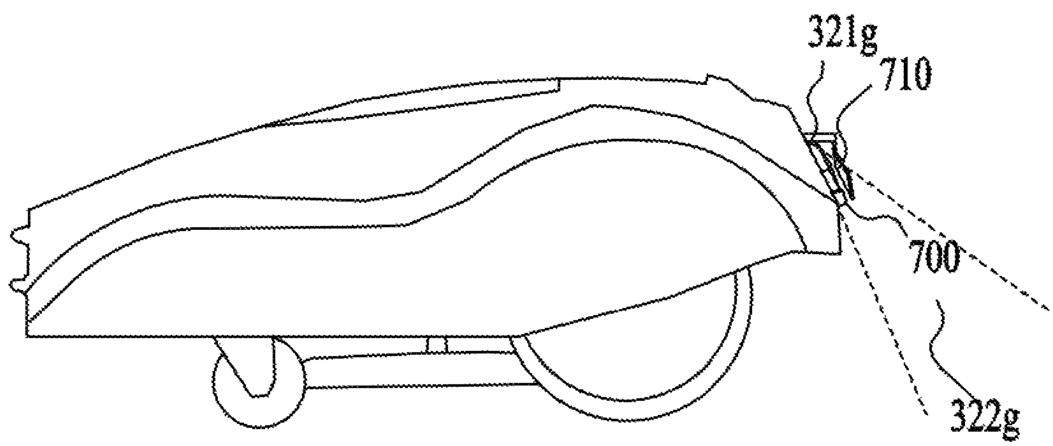


FIG. 25

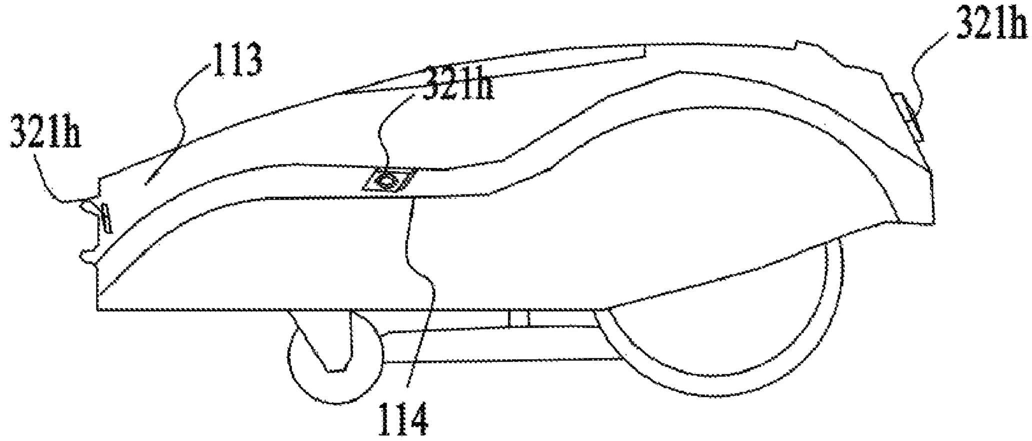


FIG. 26

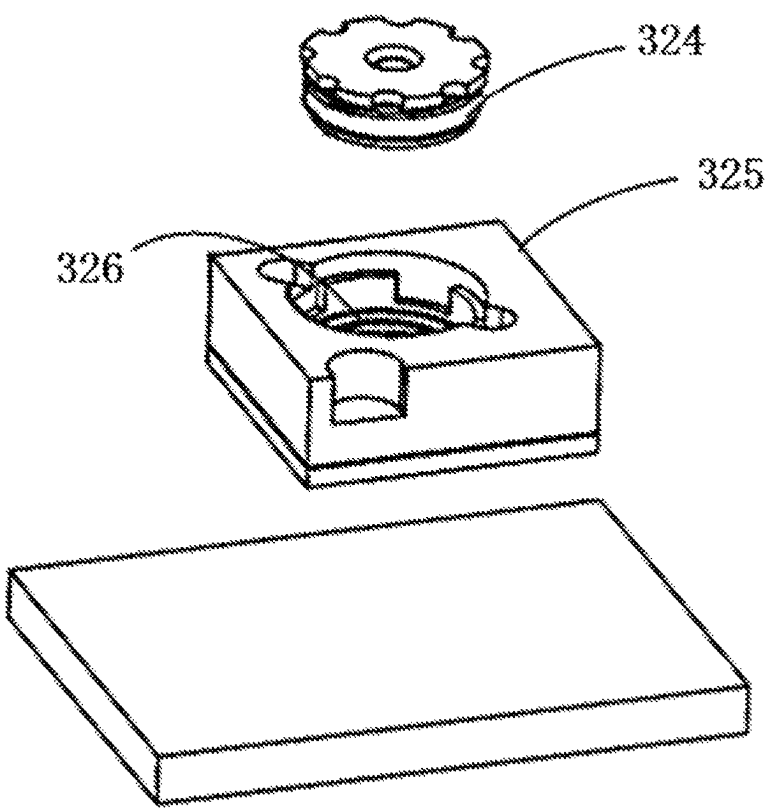


FIG. 27

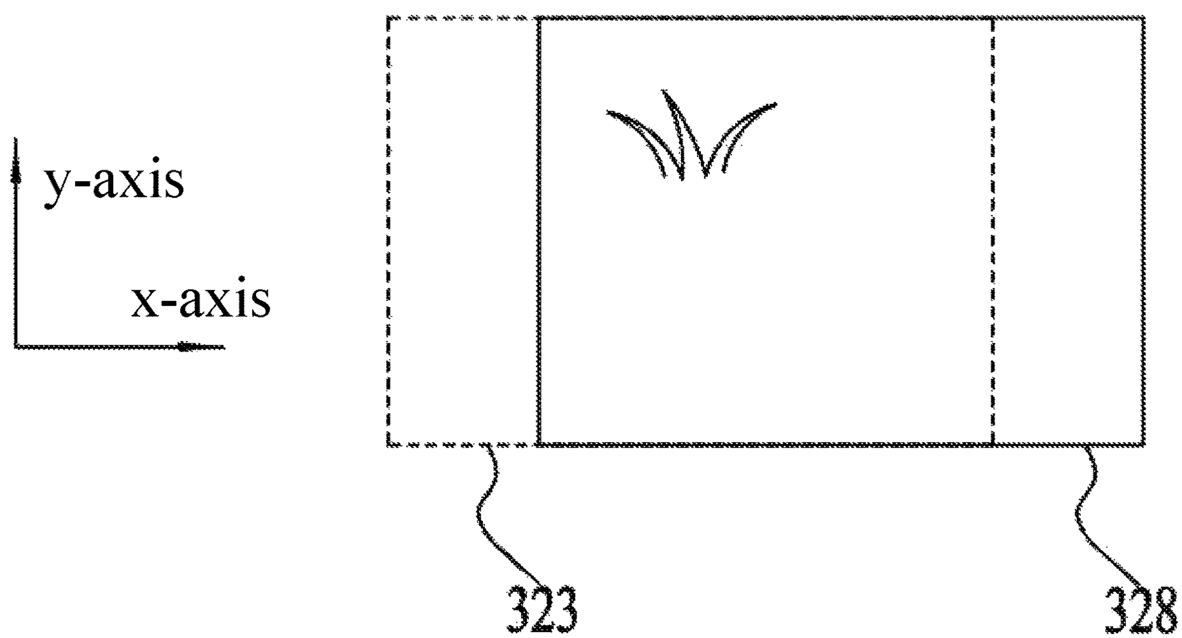


FIG. 28A

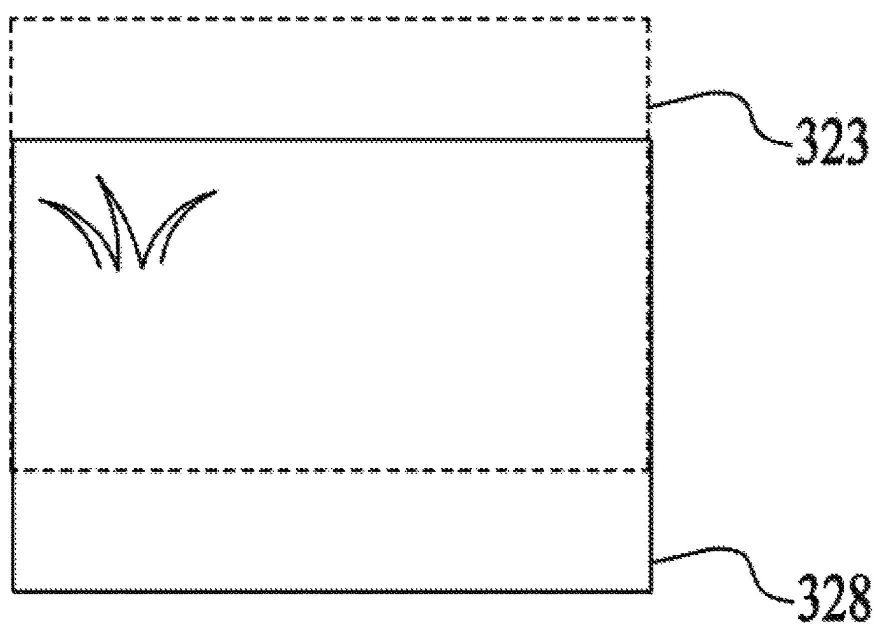


FIG. 28B

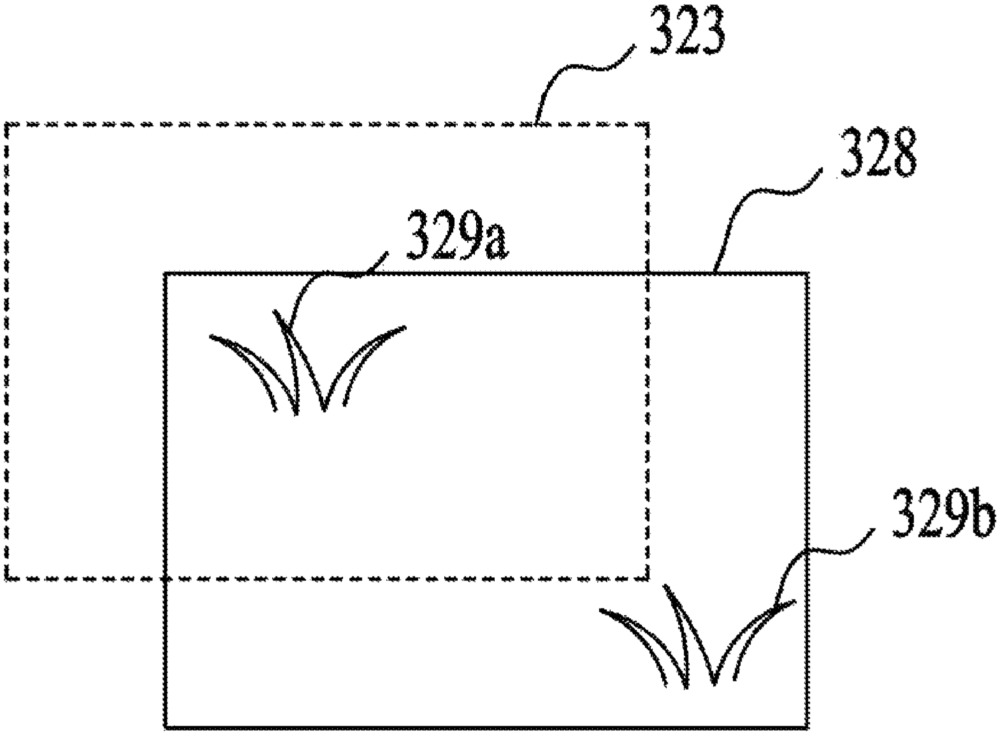


FIG. 28C

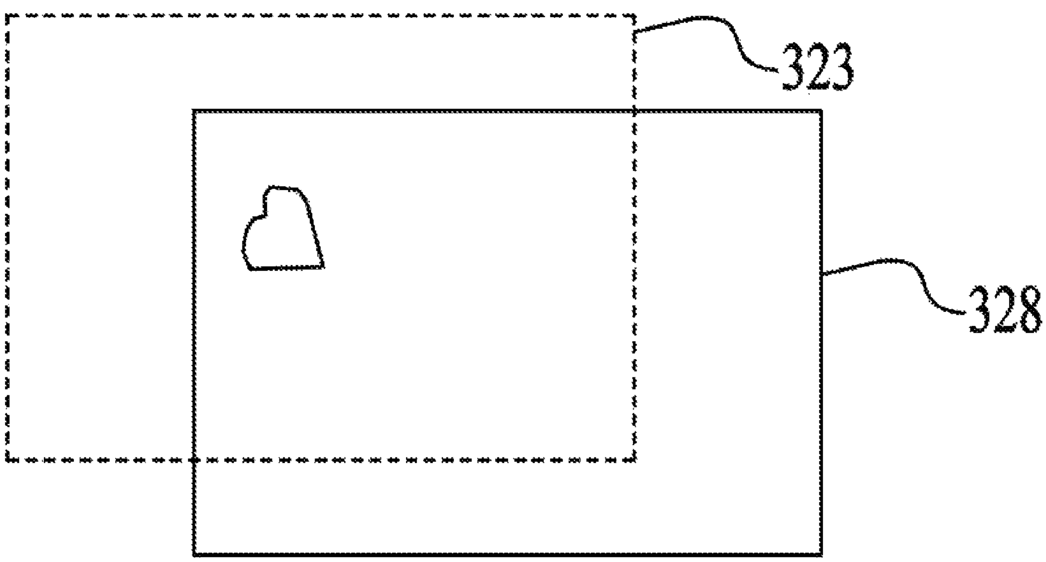


FIG. 28D

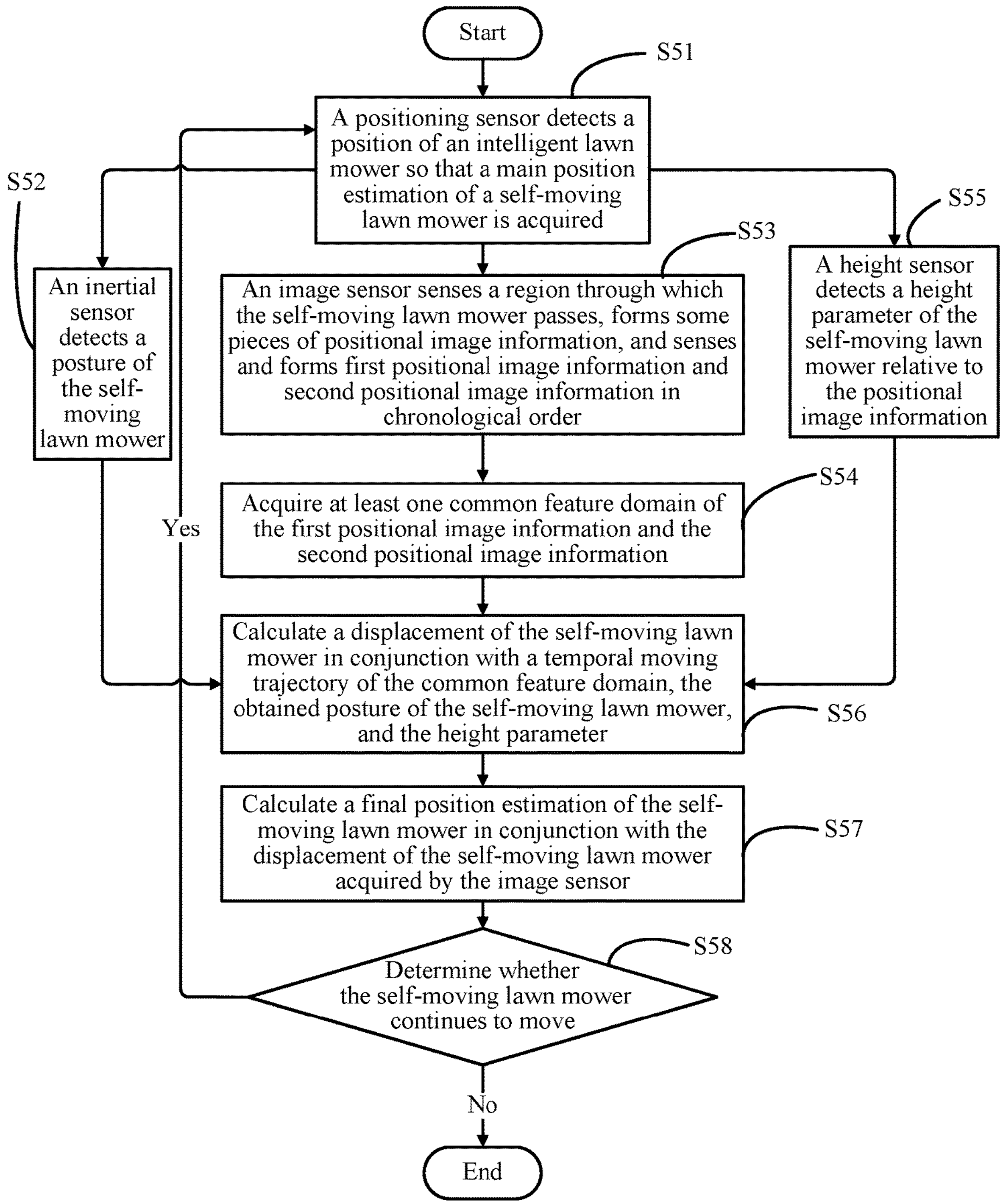

Start
S51
A positioning sensor detects a position of an intelligent lawn mower so that a main position estimation of a self-moving lawn mower is acquired
S52
An inertial sensor detects a posture of the self-moving lawn mower
S53
An image sensor senses a region through which the self-moving lawn mower passes, forms some pieces of positional image information, and senses and forms first positional image information and second positional image information in chronological order
S55
A height sensor detects a height parameter of the self-moving lawn mower relative to the positional image information
S54
Acquire at least one common feature domain of the first positional image information and the second positional image information
Yes
Calculate a displacement of the self-moving lawn mower in conjunction with a temporal moving trajectory of the common feature domain, the obtained posture of the self-moving lawn mower, and the height parameter
S56
Calculate a final position estimation of the self-moving lawn mower in conjunction with the displacement of the self-moving lawn mower acquired by the image sensor
S57
Determine whether the self-moving lawn mower continues to move
S58
No
End

FIG. 29

SELF-MOVING LAWN MOWER AND SUPPLEMENTARY OPERATION METHOD FOR AN UNMOWED REGION THEREOF

RELATED APPLICATION INFORMATION

This application is a continuation of International Application Number PCT/CN2020/111632, filed on Aug. 27, 2020, through which this application also claims the benefit under 35 U.S.C. § 119(a) of Chinese Patent Application No. 201910793036.2, filed on Aug. 27, 2019, Chinese Patent Application No. CN 201910865102.2, filed on Sep. 12, 2019, and Chinese Patent Application No. CN 201911411522.X, filed on Dec. 31, 2019, all of which are incorporated herein by reference in their entirety.

BACKGROUND

As an outdoor mowing tool, an intelligent self-moving lawn mower is favored by users due to no need for long-term operation of the users and its intelligence and convenience. At present, the route planning of the self-moving lawn mower is generally a random operation so that there will be unmowed regions. A relatively long time is needed for supplementary operation for all the unmowed regions through the random operation of the self-moving lawn mower. Therefore, the self-moving lawn mower has a relatively short service life and wastes energy, and the unmowed regions easily appear so that a lawn is not beautiful.

SUMMARY

In one example, a self-moving mowing system is provided. The self-moving mowing system includes a self-moving lawn mower. The self-moving lawn mower includes a main body, a mowing element, an output motor, moving wheels, and a drive motor. The main body includes a casing. The mowing element is connected to the main body and configured to cut vegetation. The output motor is configured to drive the mowing element. The moving wheels are connected to the main body. The drive motor is configured to drive the moving wheels to rotate. The self-moving mowing system further includes a control unit and a positioning assembly. The control unit is connected to and configured to control the output motor and the drive motor. The positioning assembly includes at least an image sensor configured to sense a region through which the self-moving lawn mower passes and form some pieces of positional image information to analyze a relative displacement of the self-moving lawn mower. The control unit includes a boundary acquisition module, an unmowed region determination module, a filling planning module, and a control module. The boundary acquisition module is configured to acquire information of an operation boundary to control the self-moving lawn mower to operate within the operation boundary. The unmowed region determination module is configured to identify an unmowed region within the operation boundary and position information of the unmowed region. The filling planning module is configured to generate an operation route for sequentially mowing in at least one unmowed region among all unmowed regions. The control module is configured to control the self-moving lawn mower to mow in the at least one unmowed region among all the unmowed regions according to the operation route.

In one example, the image sensor is configured to sense and form first positional image information and second positional image information in chronological order, where

the first positional image information and the second positional image information have at least one common feature area; the control unit is configured to obtain, through an analysis, a relative displacement of the self-moving lawn mower according to at least a temporal moving trajectory of the at least one common feature area of the first positional image information and the second positional image information of the image sensor.

In one example, a self-moving mowing system is provided. The self-moving mowing system includes a self-moving lawn mower. The self-moving lawn mower includes a main body, a mowing element, an output motor, moving wheels, and a drive motor. The main body includes a casing. The mowing element is connected to the main body and configured to cut vegetation. The output motor is configured to drive the mowing element. The moving wheels are connected to the main body. The drive motor is configured to drive the moving wheels to rotate. The self-moving mowing system further includes a control unit. The control unit is connected to and configured to control the output motor and the drive motor. The control unit includes a boundary acquisition module, an unmowed region determination module, a filling planning module, and a control module. The boundary acquisition module is configured to acquire information of an operation boundary to control the self-moving lawn mower to operate within the operation boundary. The unmowed region determination module is configured to identify an unmowed region within the operation boundary and position information of the unmowed region. The filling planning module is configured to generate an operation route for sequentially mowing in at least one unmowed region among all unmowed regions. The control module is configured to control the self-moving lawn mower to mow in the at least one unmowed region among all the unmowed regions according to the operation route.

In one example, a shortest route for sequentially mowing in the at least one unmowed region among all the unmowed regions is defined as a shortest operation route, and a ratio of a length of the operation route to a length of the shortest operation route is greater than or equal to 1 and less than or equal to 1.2.

In one example, the ratio of the length of the operation route to the length of the shortest operation route is greater than or equal to 1 and less than or equal to 1.1.

In one example, the filling planning module is configured to calculate the shortest operation route for supplementary operation for all the unmowed regions of the self-moving lawn mower.

In one example, the control module is configured to control the self-moving lawn mower to perform the supplementary operation on the unmowed regions in a sequence in which the supplementary operation is performed on the unmowed regions along the shortest operation route.

In one example, after the self-moving lawn mower performs the supplementary operation on the unmowed region, a coverage rate of the self-moving lawn mower controlled to perform the supplementary operation on the unmowed region is greater than 80%.

In one example, the positioning assembly includes one of or a combination of a global positioning system (GPS) positioning unit, an inertial measurement unit (IMU), a displacement sensor, or an image sensor.

In one example, the positioning assembly is configured to acquire an operation trajectory of the self-moving lawn mower, and a non-operated region within the operation boundary of the self-moving lawn mower is determined according to the operation trajectory of the self-moving lawn mower and operation boundary information, where in the case where an area of the non-operated region is greater than a preset value, it is determined that the non-operated region is an unmowed region.

In one example, a detection module includes an image sensor, where the image sensor is configured to acquire a two-dimensional image or a three-dimensional image within the operation boundary to acquire information of the unmowed regions of the self-moving lawn mower.

In one example, a supplementary operation method for an unmowed region of a self-moving mowing system is provided. The method includes selecting an operation region of the self-moving mowing system; starting the self-moving mowing system and positioning a self-moving lawn mower of the self-moving mowing system to acquire an operation moving trajectory and determine that a region covered by moving operation of the self-moving lawn mower is an operated region; analyzing a non-operated region in the operation region according to the operation region and the operated region and determining that a non-operated region with an area greater than a preset value is an unmowed region; planning an operation route for supplementary operation for the unmowed region according to information of the unmowed region; and controlling the self-moving lawn mower to perform the supplementary operation on at least one of all unmowed regions according to the operation route, where a ratio of a length of an operation route of the self-moving lawn mower to a length of a corresponding shortest operation route for completion of the supplementary operation for the at least one unmowed region among all the unmowed regions is greater than or equal to 1 and less than or equal to 1.2.

In one example, the supplementary operation method for an unmowed region of a self-moving mowing system further includes generating m simulation actuators, where each of the m simulation actuators is configured to randomly select one of all the unmowed regions as a starting point; selecting, by each of the m simulation actuators, a next unmowed region with a transition probability P, recording a simulation route simulated by each of the m simulation actuators after all the unmowed regions are separately selected by the m simulation actuators, setting a mark factor for the simulation route according to an information concentration function τ, and recording the above as an iterative calculation; after a number of iterations is greater than a preset value, acquiring a simulation route simulated by the m simulation actuators with most simulation times, and determining the simulation route as a shortest simulation route; and controlling the self-moving lawn mower to perform the supplementary operation on the at least one unmowed region among all the unmowed regions according to the shortest simulation route.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a plan view of a self-moving mowing system;

FIG. **2**A is a schematic diagram of a structural framework of the self-moving mowing system in FIG. **1**;

FIG. **2**B is a schematic diagram of a framework of an execution assembly of the self-moving mowing system in FIG. **1**;

FIG. **3** is a schematic diagram of an operation region of the self-moving mowing system in FIG. **1**;

FIG. **4** is a schematic diagram of a record of an operation trajectory of a self-moving lawn mower in the operation region in FIG. **3** at a first operation stage;

FIG. **5** is a schematic diagram of a record of an operation trajectory of a self-moving lawn mower in the operation region in FIG. **3** at a second operation stage;

FIG. **6** is an enlarged diagram of unmowed regions in a partial operation region in FIG. **5**;

FIG. **7** is a schematic diagram of unmowed regions in an operation region of a self-moving mowing system;

FIG. **8** is a schematic diagram of a shortest operation route along which a self-moving mowing system performs supplementary operation on the unmowed regions in FIG. **7**;

FIG. **9** is a schematic diagram of one iteration of simulated calculations of a shortest operation route by a self-moving mowing system;

FIG. **10** is a schematic diagram of iterative convergence of calculations of a shortest operation route by a self-moving mowing system;

FIG. **11** is a structural diagram of an interactive interface of a self-moving mowing system in a second example of the present application;

FIG. **12** is a schematic diagram of communication between a mobile device and a self-moving mowing system in a second example of the present application;

FIG. **13** is a schematic diagram of planning of a supplementary operation route according to a third example of the present application;

FIG. **14** is a flowchart of a method for a self-moving mowing system to perform supplementary operation on an unmowed region along an optimal route;

FIG. **15** is a flowchart of another method for a self-moving mowing system to perform supplementary operation on an unmowed region along an optimal route;

FIG. **16** is a flowchart of a method for identifying and determining an unmowed region of a self-moving mowing system;

FIG. **17** is a flowchart of another method for identifying and determining an unmowed region of a self-moving mowing system;

FIG. **18** is a flowchart of a supplementary operation method for an unmowed region of a self-moving mowing system;

FIG. **19** is a perspective view of a self-moving lawn mower;

FIG. **20** is a schematic diagram illustrating operation of an image sensor of the self-moving lawn mower in FIG. **19**;

FIG. **21**A is a diagram of structural modules of a self-moving lawn mower;

FIG. **21**B is a diagram of structural modules of an execution assembly of a self-moving lawn mower;

FIG. **22** is a plan view of a self-moving lawn mower according to an example;

FIG. **23** is a plan view of a self-moving lawn mower according to another example;

FIG. **24** is a plan view of a self-moving lawn mower according to another example;

FIG. **25** is a plan view of a self-moving lawn mower with a light source filtering device;

FIG. **26** is a plan view of a self-moving lawn mower according to another example;

FIG. **27** is a structural view of an image sensor;

FIG. **28**A is a diagram illustrating a principle of the image sensor in FIG. **27** detecting a relative displacement of a self-moving lawn mower on an x-axis;

FIG. **28**B is a diagram illustrating a principle of the image sensor in FIG. **27** detecting a relative displacement of a self-moving lawn mower on a y-axis;

FIG. **28**C is a diagram illustrating a principle of the image sensor in FIG. **27** detecting a relative displacement of a self-moving lawn mower;

FIG. **28**D is a diagram illustrating a principle of the image sensor in FIG. **27** detecting a relative displacement of a self-moving lawn mower; and

FIG. **29** is a flowchart for detection of a position of a self-moving lawn mower.

DETAILED DESCRIPTION

In an example of the present application, a self-moving mowing system is provided for intelligently mowing in a lawn and trimming the lawn. Referring to FIGS. **1** and **2**, the self-moving mowing system includes at least a self-moving lawn mower **10**. The self-moving lawn mower **10** includes a main body **100** and an execution assembly **101** connected to the main body **100**. The execution assembly **101** includes moving wheels **120** and a mowing element **130**. The mowing element **130** may be a blade for cutting vegetation. The main body **100** includes a casing **110**, and the casing **110** is configured to package and support parts such as the moving wheels **120** and the mowing element **130**. The moving wheels **120** are configured to drive the self-moving lawn mower **10** to move, and the mowing element **130** is configured to cut the vegetation. The self-moving lawn mower **10** further includes a control unit **200** configured to control an operation state of the self-moving lawn mower **10**. The self-moving lawn mower **10** further includes an output motor **131** configured to drive the mowing element **130** and a drive motor **123** connected to the moving wheels **120**, where the output motor **131** is configured to drive the mowing element **130** to rotate at a high speed to cut the vegetation, and the drive motor **123** is configured to drive the moving wheels **120** to rotate so as to drive the self-moving lawn mower **10** to move. The control unit **200** is configured to control the moving wheels **120** and the mowing element **130** to operate by controlling the output motor **131** and the drive motor **123**. The self-moving lawn mower **10** further includes a power supply device **140**. Optionally, the power supply device **140** is implemented as at least one battery pack and is connected to the self-moving lawn mower **10** through a battery pack interface on the self-moving lawn mower **10** so as to supply power to the output motor **131**, the drive motor **123**, and the control unit **200**.

The self-moving lawn mower **10** includes an output controller configured to control the output motor **131** and a drive controller configured to control the drive motor **123**. The output controller is connected to the control unit **200**, and the control unit **200** is configured to transmit an instruction to control the output motor **131** to operate through the output controller so as to control a cutting state of the mowing element **130**. The drive controller is connected to the drive motor **123** and configured to control the drive motor **123**, and the drive controller is communicatively connected to the control unit **200** so that after receiving a start instruction from a user or determining that the self-moving lawn mower **10** is started, the control unit **200** analyzes a moving route of the self-moving lawn mower **10** and transmits a moving instruction to the drive controller so as to control the drive motor **123** to drive the moving wheels **120** to move.

Referring to FIGS. **2**A and **2**B, the control unit **200** is disposed in the self-moving mowing system, configured to be a circuit board, and connected to an interactive interface **150** for receiving user instructions. The interactive interface is provided with a button for the user to input information. In an example, at least part of modules in the control unit **200** may also be installed in one or more mobile terminals; or the control unit **200** is communicatively connected to the mobile terminal, and the mobile terminal is used as an upper computer of the self-moving lawn mower **10**. The mobile terminal may be implemented as a smart mobile device such as a computer and a smart phone, and the user may control the operation of the self-moving lawn mower **10** through the mobile terminal. In an example, the self-moving lawn mower **10** transmits signals with the mobile terminal through a communication device, and the control unit **200** is configured to operate in the mobile terminal. The control unit **200** calculates and analyzes position information of the self-moving lawn mower **10** and transmits signals to control the operation of the self-moving lawn mower **10**.

The self-moving mowing system includes a detection module configured to detect the operation state of the self-moving lawn mower **10**, and the detection module includes at least a positioning assembly **300** configured to acquire a position of the self-moving lawn mower **10**. Real-time positioning data of the self-moving lawn mower **10** is analyzed so that the moving and mowing control of the self-moving lawn mower **10** is achieved. Referring to FIG. **3**, the self-moving lawn mower **10** is restricted to operate in an operation region **410**, the operation region **410** has an operation boundary **420**, and the self-moving lawn mower **10** moves within the operation boundary **420** and mows when the self-moving lawn mower **10** is controlled to be turned on. The operation boundary **420** may be set through line embedding, which is a common method in the art and will not be described in detail here. In another manner of limiting the operation boundary **420**, position coordinates of a target operation boundary **420** are selected so that the operation boundary **420** is selected, real-time position information of the self-moving lawn mower **10** is acquired, and the self-moving lawn mower **10** is adjusted and controlled so that the self-moving lawn mower **10** is restricted to operate within the operation boundary **420**.

Referring to FIGS. **2** and **4**, the positioning assembly **300** includes one of or a combination of a GPS positioning unit **310**, an IMU **320**, a displacement sensor **330**, and an image sensor **340** and is configured to acquire the position of the self-moving lawn mower **10**. In an example of the positioning assembly **300**, the positioning assembly **300** includes at least the GPS positioning unit **310** configured to acquire the position of the self-moving lawn mower **10**. The positioning assembly **300** further includes the image sensor **340** and/or the displacement sensor **330**. The image sensor **340** is disposed on the casing **110** or the main body **100** of the self-moving lawn mower **10** and configured to analyze displacement data of the self-moving lawn mower **10** by acquiring image information of a scene around the self-moving lawn mower **10**. The displacement sensor **330** may be disposed on the drive motor **123** or the moving wheel **120** and configured to acquire the displacement data of the self-moving lawn mower **10**. The position information is directly acquired through one of the GPS positioning unit **310**, the IMU **320**, the displacement sensor **330**, and the image sensor **340**, or relatively accurate position information is acquired through information obtained by multiple devices of the preceding devices in conjunction with correction. The self-moving mowing system includes a boundary acquisition module **250** configured to acquire information about the operation boundary so that the self-moving lawn mower **10** is controlled to operate within the operation boundary. The user sets the operation boundary **420** through the self-moving lawn mower or the mobile terminal so that the boundary acquisition module **250** acquires position information and distribution information of the operation boundary **420**, where the position information of the operation boundary **420** may be information about a distance of the operation boundary **420** from a current position of the self-moving mowing system or may be longitude and latitude positioning data of the operation boundary **420**. After the self-moving mowing system acquires an operation start instruction, the self-moving mowing system executes a first operation stage. At the first operation stage, the self-moving lawn mower **10** is controlled by the control unit **200** to move and mow in the operation region **410** within the operation boundary **420**. A moving trajectory of the self-moving lawn mower **10** may be disordered or the self-moving lawn mower **10** is controlled to operate according to a preset route.

Referring to FIG. **5**, the control unit **200** acquires a real-time position of the self-moving lawn mower **10** during operation. The control unit **200** includes a position acquisition module **220** and a storage module **240**. The position acquisition module **220** is communicatively connected to the positioning assembly **300** and configured to acquire a real-time displacement trajectory of the self-moving lawn mower **10**, and the storage module **240** is configured to store and mark mowed regions of the self-moving lawn mower **10** so that the real-time distribution of the mowed regions and unmowed regions can be analyzed. In the case where it is calculated that a ratio of an area of an operated region to an area of the operation region **410** is greater than a preset ratio, for example, 60%, it is determined that the first operation stage is completed, and the self-moving lawn mower **10** is controlled to end the first operation stage and enter a second operation stage. In another example, in the case where it is calculated that the ratio of the area of the operated region to the area of the operation region **410** is greater than 90%, the self-moving lawn mower **10** is controlled to enter the second operation stage. In another manner, in the case where it is calculated that the ratio of the area of the operated region to the area of the operation region **410** is greater than a preset value, the self-moving lawn mower **10** is controlled to enter the second operation stage, where the preset value may be greater than or equal to 60% and less than or equal to 90%.

FIG. **6** is an enlarged diagram of unmowed regions in a partial operation region **410**A in FIG. **5**. The control unit **200** further includes an unmowed region determination module **230** configured to acquire information about unmowed regions **430** of the self-moving lawn mower **10** during operation. The unmowed region determination module **230** acquires the operation boundary **420** and position information of the mowed regions stored in the storage module **240** and analyzes position information of regions other than the marked mowed regions in the operation region **410** within the operation boundary **420**, so as to acquire position information of missed regions when the self-moving lawn mower **10** operates at the first operation stage and store the position information of the unmowed region **430** in the storage module **240**, such as an area and coordinates of a non-operated region and distances between non-operated regions. Since a boundary of the non-operated region is acquired by tracking the displacement of the self-moving lawn mower **10**, the preceding parameter information of the non-operated region may be acquired.

In another manner of determining the unmowed region, the detection module includes the image sensor **340** configured to detect the information about the unmowed regions. The determination of the unmowed region **430** of the self-moving lawn mower **10** is acquired by the image sensor **340**. The image sensor **340** is disposed on the casing **110** or the main body **100**. When the self-moving lawn mower **10** is at the first operation stage, the image sensor **340** directly captures an image of the lawn, acquires a two-dimensional image or a three-dimensional image within the operation boundary so as to acquire areas and position information of the unmowed regions and the information about the unmowed regions of the self-moving lawn mower. The control unit stores the areas and position information of the unmowed regions in the storage module **240**, and refreshes information about the non-operated regions and stores the information about the non-operated regions in the self-moving lawn mower **10** as the self-moving lawn mower **10** moves and operates. After a ratio of a total area of the non-operated regions to the area of the operation region **410** is less than a preset value, for example, 10%, it is determined that the first operation stage is completed and controlled to end. At this time, the unmowed region determination module **230** acquires the information about the unmowed regions, that is, the position information and areas of the unmowed regions **430** and distances between unmowed regions **430**.

The control unit **200** further includes a filling planning module **210** configured to plan supplementary operation for the unmowed region **430**. The filling planning module **210** generates an operation route along which mowing is sequentially performed in at least one unmowed region among all the unmowed regions, analyzes the information about the unmowed regions **430** at the first operation stage, divides the unmowed regions **430** on which the supplementary operation needs to be performed into a first unmowed region, a second unmowed region, an N-th unmowed region, and so on, and orderly labels or sorts the first unmowed region, the second unmowed region, the N-th unmowed region, and so on, so as to generate a supplementary operation route for supplementary mowing at the second operation stage and control the self-moving lawn mower **10** to move and mow according to the generated supplementary operation route.

Referring to FIGS. **2**A and **2**B, the control unit **200** includes a control module **260**, the control module **260** is connected to the drive motor and the output motor and configured to control the drive motor and the output motor, and the control module **260** is configured to drive the self-moving lawn mower **10** to move and mow according to the supplementary operation route. Two moving wheels **120** are provided, which are a first moving wheel **121** and a second moving wheel **122**. The drive motor **123** includes a first drive motor and a second drive motor. The drive controller is connected to the first drive motor and the second drive motor. The control unit **200** controls a rotational speed of the first drive motor and a rotational speed of the second drive motor through the drive controller so as to control a moving state of the self-moving lawn mower **10**. At the second operation stage, that is, when the supplementary operation is performed on the unmowed regions of the self-moving lawn mower, the control unit **200** controls, according to the supplementary operation route, the self-moving lawn mower **10** to move and operate according to a trajectory of the supplementary operation route so that the self-moving lawn mower **10** sequentially moves and operates according to a sequence of the unmowed regions and thus the self-moving lawn mower **10** completes the supplementary mowing with high efficiency.

At the second operation stage, the positioning assembly **300** of the self-moving mowing system acquires real-time positioning of the self-moving lawn mower **10**. A moving speed and steering of the self-moving lawn mower **10** are controlled sequentially according to the position information of the unmowed regions **430** stored in the storage module **240**, and the self-moving lawn mower **10** is controlled to operate in the corresponding unmowed regions **430**. For example, at the second operation stage, the control unit **200** drives the self-moving lawn mower **10** to move toward the first unmowed region. By analyzing a distance between the current position of the self-moving lawn mower **10** and the first unmowed region, the control unit **200** causes the first moving wheel **121** and the second moving wheel **122** to rotate at different speeds through the drive controller and causes the self-moving lawn mower to steer and move toward the first unmowed region through the differential rotation of the first moving wheel **121** and the second moving wheel **122**. In another example, the self-moving lawn mower **10** includes the first moving wheel **121**, the second moving wheel **122**, and a differential mechanism between the first moving wheel **121** and the second moving wheel **122**, where the differential mechanism is used for the differential control of the first moving wheel **121** and the second moving wheel **122**, and the differential mechanism is controlled through the drive controller so that the first moving wheel **121** and the second moving wheel **122** move at different speeds, and thus the self-moving lawn mower **10** steers.

When the positioning assembly **300** detects that the position of the self-moving lawn mower **10** overlaps with the first unmowed region, it is determined that the self-moving lawn mower **10** moves to the first unmowed region and an instruction is transmitted to the output controller so that the output motor **131** drives the mowing element **130** to rotate so as to mow in the first unmowed region. The unmowed region determination module **230** acquires whether the first unmowed region is completed, and when the operation of the self-moving lawn mower **10** on the first unmowed region is completed, it is determined that the supplementary operation for the first unmowed region is completed, the control unit **200** controls the self-moving lawn mower **10** to face the second unmowed region, and the preceding steps are repeated until the operation on all the unmowed regions **430** or unmowed regions on which the supplementary operation needs to be performed among all the unmowed regions is completed. In this manner, the operation efficiency of the self-moving lawn mower **10** is improved, and the supplementary mowing is effectively performed on unmowed regions of the self-moving lawn mower **10** at an early stage so that the lawn mowed by the self-moving mowing system is more beautiful and energy is saved.

The unmowed region determination module **230** analyzes the non-operated region and analyzes the boundary of the non-operated region according to the operation trajectory of the self-moving lawn mower **10**. In the case where the area of the non-operated region is greater than a preset value, it is determined that the non-operated region is the unmowed region **430**, the unmowed regions on which the supplementary operation needs to be performed among all the unmowed regions are determined in conjunction with an operation of the user, and the self-moving lawn mower **10** is controlled to sequentially perform the supplementary operation on the unmowed regions **430**. In the case where the area of the non-operated region is less than the preset value, it is determined that the supplementary operation does not need to be performed on the non-operated region and the self-moving lawn mower **10** is controlled not to operate in the non-operated region.

In a first example of the present application, referring to FIG. **7**, the control unit establishes a positioning coordinate system so as to analyze the position of the self-moving lawn mower **10** and position information of the operation route of the self-moving lawn mower **10**. The positioning assembly establishes the positioning coordinate system on a horizontal plane, uses a certain point as an origin, such as a position of a charging station or a starting position of the self-moving lawn mower **10**, acquires the position information of the self-moving lawn mower **10** through the GPS positioning unit **310**, the IMU **320** and other devices of the positioning assembly, converts the position information of the self-moving lawn mower **10** into data on corresponding position coordinates in the positioning coordinate system, and stores the data on the corresponding position coordinates in the storage module. The positioning assembly records the operation trajectory of the self-moving lawn mower **10**, converts the operation trajectory into data on corresponding position coordinates, and stores the data on the corresponding position coordinates in the storage module. Through an analysis of trajectory data of the self-moving lawn mower **10**, the non-operated region is calculated, and whether the non-operated region is the unmowed region is determined according to the area or shape of the non-operated region. When the self-moving lawn mower **10** is controlled to perform the supplementary operation, the self-moving lawn mower **10** is controlled to enter the unmowed region according to an analysis of the real-time position information of the self-moving lawn mower **10** and the position information of the unmowed region. After the self-moving lawn mower **10** enters the unmowed region, the mowing element **130** is controlled to rotate to mow. Referring to FIG. **4**, the control unit **200** may generate a positioning coordinate system or an electronic map for analyzing the position and planned route of the self-moving lawn mower **10**. Referring to FIG. **8**, the filling planning module **210** stores an optimal route planning algorithm, analyzes, through the optimal route planning algorithm, supplementary operation routes for the unmowed regions, and analyzes a sequence in which the self-moving lawn mower **10** sequentially performs operation on the first unmowed region, the second unmowed region, and the N-th unmowed region to acquire a shortest operation route planned for the supplementary operation on the unmowed regions **430**. In this manner, the self-moving lawn mower **10** completes the supplementary operation on all the unmowed regions with the highest efficiency and the shortest distance, or the self-moving lawn mower **10** performs the supplementary operation on the unmowed regions with relatively high efficiency and a distance whose deviation from the shortest operation route is less than 50%, improving the operation efficiency of the self-moving lawn mower **10**. When the distance between unmowed regions **430** is calculated, a geometric center of the unmowed region may be selected as a mark point **431** of the unmowed region **430**, or an outermost point of the unmowed region in a certain direction is selected as the mark point **431**. Referring to FIG. **6**, for example, the westernmost points of all the unmowed regions **430** are selected as the mark points **431**, or any point in the unmowed region is selected as the mark point **431**. A distance between mark points **431** of every two unmowed regions **430** is calculated so as to calculate the distance between unmowed regions. With the mark point **431** of the unmowed region **430** as a target point, the self-moving lawn mower **10** moves toward the unmowed region **430** under the guidance of position information of the mark point **431**.

Due to the selection of different mark points **431** and the influence of obstacles, an error exists between the shortest operation route and an actual operation route along which the self-moving lawn mower **10** moves according to the shortest operation route and. In this example, a shortest route which is calculated by the filling planning module and along which the mowing is sequentially performed in the at least one unmowed region among all the unmowed regions is defined as the shortest operation route, and a ratio of a length of the actual operation route of the self-moving lawn mower **10** to a length of the shortest operation route is greater than or equal to 1 and less than or equal to 1.2. In one case, the ratio of the length of the operation route to the length of the shortest operation route is greater than or equal to 1 and less than or equal to 1.1.

In this example, the shortest operation route refers to a sequence in which the supplementary operation is performed on the unmowed regions, and different mark points **431** selected correspond to different lengths of supplementary operation routes. Therefore, an actual operation distance of the shortest operation route may be within an interval.

When the self-moving lawn mower **10** performs the supplementary operation on the unmowed regions, the control module **260** controls the self-moving lawn mower **10** to sequentially perform the supplementary operation on the unmowed regions according to a sequence in which the supplementary operation is performed on the unmowed regions along the shortest operation route. The self-moving lawn mower **10** enters the unmowed region, and the mowing element is driven to rotate to perform the mowing operation. The self-moving lawn mower **10** is controlled to move along a circle to operate or move in a zigzag shape to operate in the unmowed region or operate disorderly within a certain range. After a coverage rate of the supplementary operation the self-moving lawn mower is controlled to perform on the unmowed region is greater than 80%, it is determined that the supplementary operation on the unmowed region is completed. That is, a ratio of a coverage area of the operation of the self-moving lawn mower **10** to a total area of the unmowed region before the operation of the self-moving lawn mower **10** is greater than 80%. When the self-moving lawn mower **10** moves between the unmowed regions, the self-moving lawn mower **10** may pass through other unmowed regions. The case where the coverage rate of the operation of the self-moving lawn mower **10** on the unmowed region through which the self-moving lawn mower **10** passes is not greater than 80% is not the supplementary operation mentioned in this method example. The filling planning module is equipped with a route planning algorithm, so as to calculate and generate the operation route along which the mowing is sequentially performed in the at least one unmowed region of all the unmowed regions, and the control module controls the self-moving lawn mower to perform the supplementary operation on the unmowed regions according to the calculated operation route.

The filling planning module is equipped with the route planning algorithm, so as to calculate and generate a shortest operation route along which the mowing is sequentially performed in the at least one unmowed region of all the unmowed regions. The route planning algorithm may be a tabu search, a Dijkstra's algorithm, a fuzzy logic algorithm, an artificial potential field method, a spatial discrete method, an A* algorithm, an ant colony optimization algorithm, or other algorithms. In this manner, the shortest operation route for the unmowed regions is calculated, where the shortest operation route for the unmowed regions may be understood as a sequence of selective operation on the unmowed regions, and the sequence of selective operation on the unmowed regions can achieve fastest supplementary operation for the unmowed regions. The shortest operation route for the unmowed regions may also be the shortest route along which the self-moving lawn mower is controlled to perform the supplementary operation on the unmowed regions, so as to improve the speed of the supplementary operation.

A method for calculating the shortest operation route based on the route planning of the filling planning module **210** is provided. Simulation actuators are generated in the positioning coordinate system or the electronic map. The filling planning module **210** calculates simulation routes between the unmowed regions and store the simulation routes as a first simulation route **441**, a second simulation route **441**, an N-th simulation route **441**, and so on. The filling planning module **210** calculates simulation routes along which the simulation actuators pass through all the unmowed regions, that is, selects routes from the first simulation route, the second simulation route, and the N-th simulation route and sorts the routes so that each of the simulation routes covers all the unmowed regions, where total simulation routes include a first simulation route, a second simulation route, and an N-th simulation route, and different simulation routes correspond to different sequences of the first simulation route, the second simulation route, and the N-th simulation route. By way of example, the simulation route does not need to include all the simulation routes **441**, but cover all the unmowed regions. One of the first simulation route, the second simulation route, and the N-th simulation route is the shortest operation route **440** corresponding to the shortest route for operation on all to-be-operated regions **410**.

According to a preset ant colony optimization algorithm in the filling planning module **210**, the simulation actuators move along the first simulation route, the second simulation route, and the N-th simulation route **441** multiple times, and a mark factor is added to the simulation route **441** every time the simulation actuator moves along the simulation route **441**, where the mark factor is a pheromone, and simulation routes with different mark factor concentrations correspond to different selection probabilities of the simulation actuators.

Here, a method for analyzing the shortest operation route **440** is provided. The filling planning module **210** generates m simulation actuators. One simulation actuator is used as an example. The simulation actuator selects a certain unmowed region as a starting point, and a transition probability from an unmowed region i to an unmowed region j is P. Optionally, multiple simulation actuators may be provided. The filling planning module **210** generates multiple simulation actuators at the same time so that the simulation actuators select the unmowed regions for transition according to the transition probability P, and thus the shortest operation route is simulated and calculated.

The simulation actuators select a next unmowed region according to the transition probability. After the simulation actuators mark all the unmowed regions, the filling planning module **210** adds or volatilizes the mark factor according to the selection of the simulation routes **441** by the simulation actuators. The mark factor concentration is determined according to an information concentration function τ. The filling planning module **210** refreshes mark factors according to the information concentration function τ, increases a mark factor of the selected simulation route **441**, and volatilizes a mark factor of the unselected simulation route **441** to reduce the number of mark factors of the unselected simulation route **441**. The filling planning module **210** reads a distance of the simulation route **441** and uses a reciprocal of the distance of the simulation route **441** as a heuristic function η. The transition probability of the simulation actuator is positively correlated to the information concentration function τ and the heuristic function η.

The transition probability P of the simulation actuator from the unmowed region i to the unmowed region j is a ratio of a product of the heuristic function η and the information concentration function τ to transition probabilities P from the unmowed region i to the unmowed regions (a sum of products of heuristic functions η and information concentration functions τ). The unmowed region through which the simulation actuator has passed will be passed through with a reduced probability. The information concentration function τ is calculated from a sum of an information enhancement function and an information volatilization function. The information volatilization function has a volatilization coefficient such as 0.5. In this manner, in a single route simulation, the mark factor of the simulation route **441** through which the simulation actuator has passed is increased by $\Delta\tau$ according to the information enhancement function, a cumulative mark factor function is a sum of 50% of the previous number of mark factors and an information enhancement amount $\Delta\tau$, and the mark factor concentration of a route through which the simulation actuator has not passed is reduced by 50%.

FIG. **9** is a schematic diagram of one iteration of simulated calculations of the shortest operation route by the self-moving mowing system. After the simulation actuator completes one transition, the mark factor of the simulation route **441** is updated according to a preset algorithm, the unmowed region through which the simulation actuator has passed is changed to the operated region, each simulation actuator is assigned with a next unmowed region according to the transition probability P, and the preceding actions are repeated until all the unmowed regions have been selected for the simulation actuator to pass through. It is determined that the simulation actuator simulates one simulation route, the number of mark factors is updated according to the information concentration function τ, and after all the m simulation actuators complete the simulation routes, the information concentration functions τ are updated for the simulation routes, and the preceding actions are repeated and iterative calculations are performed until the simulation actuators repeat a certain simulation route more than A (a preset value) times or perform the iterative calculations more than B (a preset value) times under the guidance of the mark factors. For example, the simulation actuators repeat a certain simulation route more than (a preset value) 50 times or perform the iterative calculations more than 100 (a preset value) times under the guidance of the mark factors, and the most frequently selected simulation route is determined to be the shortest operation route **440**.

FIG. **10** is a schematic diagram of iterative convergence of calculations of the shortest operation route by the self-moving mowing system. Dashed points in the figure indicate the distance statistics of simulation operation routes of all the simulation actuators of the self-moving mowing system, and a solid line indicates the distance statistics of a simulation operation route of one simulation actuator of the self-moving mowing system in this example. After **80** iterations, the shortest operation route for the supplementary operation for the unmowed regions in the current operation situation is obtained.

In an example, the information enhancement amount $\Delta\tau$ is a ratio of a constant to a distance of the current simulation route **441**, that is, the information enhancement amount $\Delta\tau$ is calculated through an ant-cycle system. In the case where the simulation route **441** is not selected for the simulation actuator to pass through, the information enhancement amount $\Delta\tau$ is 0. The information enhancement amount $\Delta\tau$ may also be calculated through an ant-quantity system or an ant-density system, which will not be described in detail here.

The transition probability P may be calculated by the following formula:

$$P_{ij}^{k}(t)=\begin{cases}\dfrac{[\tau_{ij}(t)]^{\alpha}\cdot[\eta_{ij}(t)]^{\beta}}{\sum\limits_{s\in J_k(i)}[\tau_{is}(t)]^{\alpha}\cdot[\eta_{is}(t)]^{\beta}} & \text{if } j\in J_k(i)\\ 0 & \text{if not}\end{cases} \tag{1}$$

where α denotes a mark factor index, β denotes a heuristic index, and J_k denotes an unmowed region.

The information concentration function τ and the information enhancement amount $\Delta\tau$ are respectively expressed by the following formulas:

$$\tau_{ij}(t+n)=(1-\rho)\cdot\tau_{ij}(t)+\Delta\tau_{ij}(t) \tag{2}$$

$$\Delta\tau_{ij}(t)=\sum_{k=1}^{m}\Delta\tau_{ij}^{k}(t) \tag{3}$$

where ρ denotes a mark factor volatilization coefficient.

In another example, the optimal route planning algorithm in the filling planning module **210** adopts the A* algorithm, and the simulation actuator is generated in the positioning coordinate system or the electronic map. The filling planning module **210** calculates the simulation routes for the unmowed regions, and calculates one of the first simulation route, the second simulation route, and the N-th simulation route as the shortest operation route **440** corresponding to the shortest route for the operation on all the to-be-operated regions **410**. The filling planning module **210** establishes a heuristic cost function F=G+H, where F denotes a total movement cost, G denotes a movement cost from a parent node to a current block, and H denotes a movement cost from the current block to an end point. A solution with the smallest cost is selected for each step so that the preceding shortest operation route is calculated.

When an obstacle exists between two unmowed regions, the operation route planning of the filling planning module **210** for the supplementary operation may adopt the artificial potential field method: a potential field function is established with a target point as a gravitational force and the obstacle as a repulsive force; or a dynamic window approach (DWA) method is adopted so as to obtain a shortest simulation route for the unmowed regions **430**. In conjunction with the sequence in which the supplementary operation is performed on the unmowed regions **430** along the shortest operation route, a shortest operation distance of the shortest simulation route is obtained.

In a second example of the present application, referring to FIGS. **11** and **12**, the self-moving mowing system is provided with the interactive interface **150** for interaction with the user. The interactive interface **150** is communicatively connected to the control unit, may display operation state information of the self-moving mowing system, and is provided with buttons or switches for the user to control the start and operation of the self-moving mowing system. The interactive interface **150** is connected to the control unit. When the user transmits a control instruction through the buttons or switches, the control unit acquires and analyzes the control instruction and outputs a corresponding control instruction to a corresponding controller so as to control the operation of the self-moving mowing system. The interactive interface **150** may be installed in a mobile terminal **20** and provide the user with the operation state information of the self-moving mowing system through the mobile terminal **20**, and the user transmits the control instruction through the mobile terminal **20** to control the operation state of the self-moving mowing system.

The control unit generates an electronic map **151** to analyze a position of a self-moving lawn mower **10***a* and position information of an operation route of the self-moving lawn mower **10***a*. The electronic map may be displayed to the user through the interactive interface **150**. The positioning assembly **300** acquires a real-time position and an operation trajectory of the self-moving lawn mower **10***a*, generates a corresponding virtual self-moving lawn mower **10***a* and the operation trajectory on the electronic map **151**, analyzes a corresponding non-operated region, that is, an unmowed region **430***a* as the self-moving lawn mower **10***a* is operating, and generates a corresponding virtual unmowed region on the electronic map **151**. The control unit generates the electronic map **151** and stores information content of the electronic map **151** in the storage module so that the electronic map **151** is provided in the control unit to operate. The control unit matches an operation region **410***a* of the self-moving lawn mower **10***a* to the electronic map **151** so that an operation boundary **420***a* of the operation region **410***a* of the self-moving lawn mower **10***a* is selected through an operation on the electronic map **151** and displayed in the electronic map **151**. The control unit includes an electronic map control. The electronic map control includes world electronic map information or regional electronic map information or can load one of or a combination of online electronic map information and offline electronic map information. The electronic map control is configured to be capable of extracting the regional electronic map information. It should be understood by those skilled in the art that any positional point displayed in the electronic map **151** corresponds to international standard longitude and latitude data information, and longitude and latitude data information of any positional point in a regional map displayed in the electronic map **151** is stored in the electronic map control or on a cloud, that is, longitude and latitude data of an actual position corresponding to a positional point displayed in the electronic map **151** may be acquired through the electronic map control. The electronic map control is implemented as a G-Map control. The G-Map control is a loading processing tool of the electronic map **151** in the related art. The display content and operation data information of the electronic map **151** are initialized and loaded and display information of the electronic map **151** such as a display center point, a zoom level, a resolution, and a view type are set through the G-Map control. The boundary acquisition module and the electronic map control may be communicatively connected. The user selects the boundary of the operation region on the electronic map **151** and corresponds the boundary of the operation region to the boundary of the actual operation region so that the boundary acquisition module acquires the operation boundary of a region to be operated.

The unmowed region determination module analyzes the non-operated region according to the operation trajectory of the self-moving lawn mower **10***a*, and the filling planning module determines whether the non-operated region is the unmowed region **430***a*, projects the region to be operated on the electronic map **151**, generates the virtual unmowed region on the electronic map **151**, analyzes the boundary of the unmowed region, generates the boundary of the virtual unmowed region on the electronic map **151**, and determines whether the non-operated region is the unmowed region. When the positioning assembly **300** detects that the position of the virtual self-moving lawn mower **10***a* is within the boundary of the virtual unmowed region, that is, the positioning assembly **300** detects that the position of the self-moving lawn mower **10***a* is within the boundary of the unmowed region, it is determined that the self-moving lawn mower **10***a* moves to the first unmowed region and an instruction is transmitted to the output controller so that the output motor **131** drives the mowing element **130** to rotate so as to mow in the first unmowed region. The self-moving lawn mower **10***a* operates in the unmowed region **430***a*. The unmowed region determination module acquires whether the first unmowed region is completed, and when the operation of the self-moving lawn mower **10***a* on the first unmowed region is completed, it is determined that the supplementary operation for the first unmowed region is completed, the control unit controls the self-moving lawn mower **10***a* to face the second unmowed region, and the preceding steps are repeated until the operation on all unmowed regions **430***a* is completed.

The control unit correspondingly displays simulated unmowed regions through the electronic map **151**, that is, the simulated unmowed regions correspond to the actual unmowed regions in position, and the electronic map **151** is displayed through the interactive interface so that the user can acquire a state of the unmowed region. Through the interactive interface, the user may selectively output information to add or delete the unmowed region. Exemplarily, the user observes a situation within the operation boundary to analyze a situation of the unmowed region, such as an operated region that actually needs to be added or whether an obstacle exists in the operation region, and the user analyzes and determines a position of the unmowed region on the electronic map **151** or in the positioning coordinate system, and selects a corresponding unmowed region on the electronic map **151** or in the positioning coordinate system through the interactive interface to add the unmowed region or delete the corresponding unmowed region selected by the system. The filling planning module generates a simulation actuator a configured to move and operate in the unmowed regions.

In a third example of the present application, the difference from the first example is that at the second operation stage, that is, when the supplementary operation is performed on the unmowed regions of the self-moving lawn mower, the control module is configured to control the self-moving lawn mower to mow in the at least one unmowed region among all the unmowed regions one by one. Referring to FIG. **13**, the filling planning module plans a supplementary route **440***b* for unrepeated operation on the unmowed regions according to positions of unmowed regions **430***b*, and the control unit controls, according to a sequence in which the supplementary operation is performed on the unmowed regions **430***b*, the self-moving lawn mower to move and operate according to a trajectory of the supplementary operation route. In this manner, the self-moving lawn mower sequentially moves and operates in the sequence of the unmowed regions so that the self-moving lawn mower performs the unrepeated operation on the unmowed regions and completes the supplementary mowing with high efficiency.

At the second operation stage, the positioning assembly of the self-moving mowing system acquires real-time positioning of the self-moving lawn mower. A moving speed and steering of the self-moving lawn mower are controlled sequentially according to position information of the unmowed regions **430***b* stored in the storage module, and the self-moving lawn mower is controlled to operate in the corresponding unmowed regions **430***b*, where the unmowed regions **430***b* in the figure may also be understood as mark points. For example, at the second operation stage, the control unit drives the self-moving lawn mower to move toward a certain unmowed region. By analyzing a distance between the current position of the self-moving lawn mower and the first unmowed region, the control unit causes the first moving wheel and the second moving wheel to rotate at different speeds through the drive controller and causes the self-moving lawn mower to steer and move toward the first unmowed region through the differential rotation of the first moving wheel and the second moving wheel.

When the positioning assembly detects that the position of the self-moving lawn mower overlaps with the first unmowed region, it is determined that the self-moving lawn mower moves to the first unmowed region and an instruction is transmitted to the output controller so that the output motor drives the mowing element to rotate so as to mow in the first unmowed region. The unmowed region determination module acquires whether the first unmowed region is completed, and when the operation of the self-moving lawn mower in the first unmowed region is completed, it is determined that the supplementary operation for the first unmowed region is completed, the control unit controls the self-moving lawn mower to face the second unmowed region, and the preceding steps are repeated until the operation on all the unmowed regions **430***b* is completed and mowing is not repeated in each unmowed region. In this manner, the operation efficiency of the self-moving lawn mower is improved, and the supplementary mowing is effectively performed on unmowed regions of the self-moving lawn mower at an early stage so that the lawn mowed by the self-moving mowing system is more beautiful and energy is saved. In a manner of planning the supplementary operation route for the unmowed regions, the filling planning module is equipped with an algorithm, a point closest to the self-moving lawn mower is selected as a starting point, and an unmowed region that is closest to a previous unmowed region or the starting point is selected as a next unmowed region until all the unmowed regions are selected, and the corresponding supplementary operation route is generated according to a sequence in which the unmowed regions are selected. Therefore, the control module is configured to: after the self-moving lawn mower is controlled to perform the supplementary operation on the N-th unmowed region, select an M-th unmowed region on which the supplementary operation is not performed and which has a smallest distance from the N-th unmowed region as a next unmowed region on which the supplementary operation is performed. The self-moving lawn mower is controlled, according to the supplementary operation route, to perform the supplementary operation on the unmowed regions in the corresponding sequence to complete the supplementary operation on all the unmowed regions and the supplementary operation for each unmowed region is not repeated, thereby improving the operation efficiency of the self-moving lawn mower.

Referring to FIG. **14**, a method for the self-moving lawn mower **10** to perform the supplementary operation on the unmowed region **430** along an optimal route is provided. In S**1**, the position information and area of the non-operated region of the self-moving lawn mower **10** are acquired, if the area of the non-operated region is greater than a preset value L, it is determined that the non-operated region is the unmowed region, and the distances between the unmowed regions are calculated. In S**2**, the m simulation actuators are generated, where each simulation actuator randomly selects an unmowed region as the starting point. In S**3**, each of the m simulation actuators selects the next unmowed region with the transition probability P, where if an unmowed region has been selected by a simulation actuator n, the selected unmowed region is selected by the simulation actuator n with a reduced probability. In S**4**, whether all the unmowed regions are separately selected by the m simulation actuators is determined. If not, step S**3** is repeated. If so, step S**5** is performed. In S**5**, a simulation route simulated by each of the m simulation actuators is recorded, a mark factor is set for the simulation route according to the information concentration function τ, and the above is recorded as an iterative calculation. In S**6**, whether the number of iterations is greater than a preset value is determined. If not, step S**2** is repeated. If so, step S**7** is performed. In S**7**, a simulation route simulated most frequently by m self-moving lawn mowers **10** is acquired and determined to be the shortest operation route **440**. In S**8**, the self-moving lawn mower **10** is controlled to perform the supplementary operation on the at least one unmowed region among all the unmowed regions according to the shortest simulation route.

Referring to FIG. **15**, a method for the self-moving lawn mower **10** to perform the supplementary operation on the unmowed region **430** along an optimal route is provided. In S**11**, the position information and area of the non-operated region of the self-moving lawn mower **10** are acquired, if the area of the non-operated region is greater than a preset value L, it is determined that the non-operated region is the unmowed region, and the distances between the unmowed regions are calculated. In S**12**, the m simulation actuators are generated, where each simulation actuator randomly selects an unmowed region as the starting point. In S**13**, each of the m simulation actuators selects the next unmowed region with the transition probability P, where if an unmowed region has been selected by a simulation actuator n, the selected unmowed region is selected by the simulation actuator n with a reduced probability. In S**14**, whether all the unmowed regions are separately selected by the m simulation actuators is determined. If not, step S**3** is repeated. If so, step S**15** is performed. In S**15**, a simulation route simulated by each of the m simulation actuators is recorded, a mark factor is set for the simulation route according to the information concentration function τ, and the above is recorded as an iterative calculation. In S**16**, it is determined whether the number of times a simulation route is simulated is greater than a preset value N. If not, steps S**2** to S**4** are repeated. If so, step S**17** is performed. In S**17**, the simulation route is acquired and determined to be the shortest operation route. In S**18**, the self-moving lawn mower **10** is controlled to perform the supplementary operation on the at least one unmowed region among all the unmowed regions according to the shortest simulation route. To sum up, referring to FIG. **16**, a method for identifying and determining the unmowed region **430** of the self-moving lawn mower **10** is provided. In S**21**, the operation region **410** of the self-moving lawn mower **10** is selected. In S**22**, the self-moving lawn mower **10** is started, the positioning assembly **300** acquires and records the operation moving trajectory of the self-moving lawn mower **10**, and it is determined that a region through which the self-moving lawn mower **10** operates is the mowed region. In S**23**, whether the ratio of the area of the operated region of the self-moving lawn mower **10** to the total area of the operation region **410** is greater than a preset value M is calculated. In S**24**, if so, the non-operated region of the self-moving lawn mower **10** in the operation region **410** is acquired and determined to be the unmowed region **430**. Optionally, the area of each non-operated region of the self-moving lawn mower **10** is calculated, where if the area of the non-operated region is greater than the preset value L, it is determined that the non-operated region is the unmowed region **430**.

Referring to FIG. **17**, in another example, a method for identifying and determining the unmowed region **430** of the self-moving lawn mower **10** is provided. In S**31**, the operation region **410** of the self-moving lawn mower **10** is selected. In S**32**, the self-moving lawn mower **10** is started, and the image sensor **340** acquires and updates the non-operated region of the intelligent self-moving lawn mower **10**. In S**33**, whether a ratio of an area of a current non-operated region of the self-moving lawn mower **10** to the total area of the operation region **410** is less than a preset value N is determined. In S**34**, if so, the non-operated region currently analyzed by the image sensor **340** is acquired and determined to be the unmowed region **430**. The area of each non-operated region of the self-moving lawn mower **10** is calculated, where if the area of the non-operated region is greater than the preset value L, it is determined that the non-operated region is the unmowed region **430**.

Referring to FIG. **18**, a supplementary operation method for the unmowed region **430** of the self-moving lawn mower **10** is provided. In S**41**, the operation region of the self-moving mowing system is selected. In S**42**, the self-moving mowing system is started, the self-moving lawn mower **10** of the self-moving mowing system is positioned so that the operation moving trajectory is acquired, and it is determined that a coverage region of moving operation of the self-moving lawn mower **10** is the operated region. In S**43**, the non-operated region in the operation region is analyzed according to the operation region and the operated region, and it is determined that the non-operated region with an area greater than the preset value is the unmowed region. In S**44**, if so, the non-operated region in the operation region is analyzed according to the operation region and the operated region, it is determined that the non-operated region with an area greater than the preset value is the unmowed region, and the operation route for the supplementary operation on the unmowed regions is planned according to the information about the unmowed regions; if not, step S**43** is repeated. In S**45**, S**44** is repeated until the ratio of the length of the operation route of the self-moving lawn mower **10** to the length of the corresponding shortest operation route for the at least one unmowed region among all the unmowed regions is greater than or equal to 1 and less than or equal to 1.2.

In an example of the present application, to position the self-moving lawn mower to determine the position information of the unmowed regions of the self-moving lawn mower, a self-moving lawn mower is provided. Referring to FIGS. **19** to **21**, the self-moving lawn mower includes at least a main body **100***a* and an execution assembly **101***a* connected to the main body **100***a*. The execution assembly **101***a* includes a mowing element **110***a* and an output motor **120***a* configured to drive the mowing element **110***a* to rotate. The mowing element **110***a* is disposed below the main body **100***a*. The mowing element **110***a* is driven to rotate by the output motor **120***a* and configured to cut vegetation. The execution assembly **101***a* includes moving wheels **130***a* and a drive motor **140***a* configured to provide a driving force to the moving wheels **130***a* for the moving wheels **130***a* to rotate. The self-moving lawn mower further includes a control unit **200***a*. The control unit **200***a* controls rotational speeds of the moving wheels **130***a* by controlling the drive motor **140***a* so as to control a moving state of the self-moving lawn mower. The self-moving lawn mower further includes a power supply device **800**. Optionally, the power supply device **800** is implemented as at least one battery pack and is connected to the self-moving lawn mower through a battery pack interface on the self-moving lawn mower so as to supply power to the output motor **120***a* and the drive motor **140***a*. The self-moving lawn mower includes an output controller **150***a* configured to control the output motor **120***a* and a drive controller **160***a* configured to control the drive motor **140***a*. The output controller **150***a* is connected to the control unit **200***a*, and the control unit **200***a* is configured to transmit an instruction to control the output motor **120***a* to operate through the output controller **150***a* so as to control a cutting state of the mowing element **110***a*. The drive controller **160***a* is connected to the drive motor **140***a* and configured to control the drive motor **140***a*, and the drive controller **160***a* is communicatively connected to the control unit **200***a* so that after receiving a start instruction from a user or determining that the self-moving lawn mower is started, the control unit **200***a* analyzes a moving route of the self-moving lawn mower and transmits a moving instruction to the drive controller **160***a* so as to control the drive motor **140***a* to drive the moving wheels **130***a* to move.

The self-moving lawn mower is provided with a positioning assembly **300***a* configured to detect a position of the self-moving lawn mower and an information collection module **400***a* configured to collect position information of the self-moving lawn mower. The information collection module **400***a* is configured to determine a current mowed route of the lawn mower by acquiring the position information of the self-moving lawn mower detected by the positioning assembly **300***a*. The information collection module **400***a* is connected to the control unit **200***a* and configured to transmit the position information of the self-moving lawn mower to the control unit **200***a*. The positioning assembly **300***a* includes at least a first positioning unit **310***a* and a second positioning unit **320***a*. The first positioning unit **310***a* includes a positioning sensor **311**. The positioning sensor **311** may be a GPS positioning assembly or a global navigation satellite system (GNSS) positioning assembly and configured to acquire primary position information or a main position estimation and a starting position of movement of the self-moving lawn mower. The second positioning unit **320***a* is configured to detect accurate position information of the self-moving lawn mower, that is, based on the primary position information acquired by the first positioning unit **310***a*, the second positioning unit **320***a* determines the accurate position information with a smaller error to acquire a final position estimation of the self-moving lawn mower. The first positioning unit **310***a* and the second positioning unit **320***a* transmit the detected position information to the information collection module **400***a* for the information collection module **400***a* to analyze a displacement state of the self-moving lawn mower. The GNSS positioning assembly is a global navigation satellite system, which includes GPS of the US, Beidou of China, GLONASS of Russia, GALILEO of the European Union, regional systems such as the Quasi-Zenith Satellite System (QZSS) of Japan and the Indian Regional Navigation Satellite System (IRNSS) of India, and augmentation systems such as WASS of the US, the Multi-functional Satellite Augmentation System (MSAS) of Japan, the European Geostationary Navigation Overlay Service (EGNOS) of the European Union, the

GPS-aided GEO augmented navigation (GAGAN) of India, and NIG-GOMSAT-1 of Nigeria.

The second positioning unit **320***a* includes at least an image sensor **321**. The image sensor **321** is disposed on the main body **100***a*. The image sensor **321** generates a corresponding image sensing region **322** and acquires an image in the image sensing region **322**. The self-moving lawn mower moves to drive the image sensor **321** to move so that the image sensing region **322** moves so as to acquire different images. The image sensor **321** continuously or intermittently acquires images in the image sensing region **322** and acquires first positional image information **323** at a first time point and second positional image information **328** at a second time point thereafter. At least one common feature area of the first positional image information **323** and the second positional image information **328** is acquired. The common feature area includes one or more feature points. Displacement states of one or more common feature points are compared so that the displacement state or relative displacement state of the self-moving lawn mower on the ground is acquired. In an example of the present application, the common feature area may be a collection of points, lines, surfaces, or colors in the image, such as a histogram of oriented gradients (HOG) feature, an Hddr feature, and a Large Electron-Positron Collider (LEP) feature, and an actual displacement of the self-moving lawn mower within the corresponding time is analyzed through a temporal moving trajectory of the common feature area.

For example, referring to FIG. **28**A, assuming that the self-moving lawn mower has a relative position only on an x-axis, a relative displacement of a first grass **329***a* in the first positional image information **323** and the second positional image information **328** on the x axis is detected so that a displacement of the self-moving lawn mower on the x-axis can be acquired. Referring to FIG. **28**B, assuming that the self-moving lawn mower has a relative position only on the y-axis, a relative displacement of the first grass **329***a* in the first positional image information **323** and the second positional image information **328** on the y-axis is detected so that a displacement of the self-moving lawn mower on the y-axis can be acquired. Referring to FIG. **28**C, assuming that the self-moving lawn mower has relative positions on the x-axis and the y-axis, a vegetation region A is the first positional image information **323**, and a vegetation region A' is the second positional image information **328**. Through a relative positional relationship between the grass **329***a* in the first positional image information **323** and the grass **329***a* in the second positional image information **328**, the relative positions of the self-moving lawn mower on the x-axis and the y-axis are determined, and then the position information of the self-moving lawn mower is acquired. The control unit **200***a* or the information collection module **400***a* may establish a coordinate system to calculate the position of the self-moving lawn mower, where a plane where the x-axis and the y-axis are located is parallel to a horizontal plane. Referring to FIG. **28**D, the first positional image information and the second positional image information may also be image information of other objects such as obstacles.

The image sensor is configured to detect the relative displacement of the self-moving lawn mower. The image sensor continuously or intermittently acquires the moving trajectory of one or more common feature areas in the first positional image information and the second positional image information **328** of the vegetation behind the moving wheels through an imaging device of the image sensor so as to acquire the position information of the self-moving lawn mower or the relative displacement of the self-moving lawn mower on the x-axis and/or the y-axis. For example, the imaging device of the image sensor projects light onto the vegetation near a rear side of the moving wheels. With the moving time and moving displacement of the self-moving lawn mower, within a first time period to, the moving trajectory of the first grass **329***a* in the vegetation region A and the vegetation region A' on a film of the image sensor is acquired, that is, the relative displacement of the first grass **329***a* is obtained so as to acquire the relative displacement of the self-moving lawn mower within the first time period, where the vegetation region A and the vegetation region A' have an overlapping region and the first grass **329***a* is located at different positions in the overlapping region of the vegetation region A and the vegetation region A'. Within the following second time period t**1**, the moving trajectory of a second grass **329***b* in a vegetation region B and a vegetation region B' on the film of the image sensor is acquired, that is, the relative displacement of the second grass **329***b* is obtained so as to acquire the relative displacement of the self-moving lawn mower within the second time period, where the vegetation region B and the vegetation region B' have an overlapping region and the second grass **329***b* is located at different positions in the overlapping region of the vegetation region B and the vegetation region B'. Exemplarily, when the self-moving lawn mower moves on the ground with time passing, if the self-moving lawn mower and the image sensor (including the film) installed on the self-moving lawn mower are assumed to be stationary, the image sensor acquires the relative displacement state of the first grass **329***a* or the second grass **329***b* in the first positional image information and the second positional image information. In this manner, the displacement information of the first grass **329***a* and the second grass **329***b* in any time segment is accumulated and superimposed so that the displacement information of the self-moving lawn mower on the ground is obtained.

An acquisition interval of the first positional image information **323** and the second positional image information **328** may be set according to a speed of the self-moving lawn mower so that one common feature area of the first positional image information **323** and the second positional image information **328** can be effectively acquired, and thus the moving displacement information of the self-moving lawn mower is determined through the common feature area.

Optionally, the second positioning unit **320***a* may further include the image sensor and a calibration device **327**. The calibration device **327** is configured to detect posture information and/or position information of the self-moving lawn mower to compensate for an offset error of a calculation according to the positional image information of the image sensor. The calibration device **327** may include an inertial sensor **325**. The image sensor on the self-moving lawn mower acquires the moving trajectory of the self-moving lawn mower, and the inertial sensor **325** acquires angular offset information on the moving trajectory of the self-moving lawn mower so as to correct the position information obtained by the image sensor and obtain more accurate position information of the self-moving lawn mower.

The inertial sensor detects a posture of the self-moving lawn mower, and the control unit calculates a calibration coefficient according to the posture of the self-moving lawn mower and calibrates, according to the calibration coefficient, the displacement of the self-moving lawn mower obtained by the positional image information.

The control unit is configured to obtain, through an analysis, the relative displacement on the x-axis and/or the y-axis or the position information of the self-moving lawn mower according to the temporal moving trajectory of at least one common feature area of the first positional image information and the second positional image information of the image sensor. The inertial sensor can detect an angle of the main body relative to the plane where the x-axis and the y-axis are located.

The control unit is configured to obtain, through the analysis, the relative displacement on the x-axis and/or the y-axis or the position information of the self-moving lawn mower according to the temporal moving trajectory of the at least one common feature area of the first positional image information and the second positional image information of the image sensor. A height sensor can measure a distance between the self-moving lawn mower relative to an object corresponding to the common feature area on a z-axis. The control unit is configured to obtain, through the analysis, the relative displacement on the x-axis and/or the y-axis or the position information of the self-moving lawn mower according to the distance between the self-moving lawn mower and the object corresponding to the common feature area on the z-axis, the angle of the main body relative to the plane where the x-axis and the y-axis are located, and the temporal moving trajectory of the at least one common feature area of the first positional image information and the second positional image information of the image sensor.

The inertial sensor **325** may be implemented as an IMU. The IMU includes an accelerometer and a gyroscope and is configured to detect the angular offset information of the self-moving lawn mower in a moving process so as to determine the posture of the self-moving lawn mower. The control unit calculates the calibration coefficient according to the posture of the self-moving lawn mower and calibrates, according to the calibration coefficient, the displacement of the self-moving lawn mower obtained by the positional image information. The inertial sensor is applied to the acquisition of the posture of the self-moving lawn mower while the image sensor acquires the first positional image information and the second positional image information for the analysis and calibration of the first positional image information and the second positional image information.

The inertial sensor **325** transmits the detected angular offset information of the self-moving lawn mower to the information collection module **400***a*, and the information collection module **400***a* analyzes final displacement information of the self-moving lawn mower according to the detected primary position information, displacement information, and angular offset information of the self-moving lawn mower so as to acquire real-time accurate position data of the self-moving lawn mower.

To avoid an error of detection of the positioning assembly **300***a* or the image sensor due to the undulating terrain of the lawn, the calibration device **327** may further include a height sensor **326** configured to detect level height information of the lawn. The height sensor **326** may also be a distance sensor configured to measure a distance between the self-moving lawn mower and an object to be measured in a height direction. For example, the height sensor **326** may be a distance measuring sensor such as a time-of-flight (TOF) sensor. The TOF sensor emits light and calculates a height of the lawn relative to the image sensor by calibrating a difference between when the light is emitted and when the light is reflected by an object such as the lawn or the ground. The height sensor **326** may detect the distance between the self-moving lawn mower and the object corresponding to the common feature area on the z-axis. For example, if the common feature area is vegetation, the height sensor **326** can measure the distance between the vegetation and the self-moving lawn mower. The control unit calculates the calibration coefficient according to a height parameter of the self-moving lawn mower and calibrates, according to the calibration coefficient, the displacement of the self-moving lawn mower obtained by the positional image information, thereby improving the accuracy of the displacement of the self-moving lawn mower detected by the image sensor.

The height sensor **326** transmits the detected level height data of the lawn relative to the image sensor to the information collection module **400***a*, and the information collection module **400***a* generates a calibration coefficient for the position information detected by the positioning assembly **300***a* according to acquired real-time height information of the self-moving lawn mower. The position information detected by the image sensor **321** is adjusted by using the calibration coefficient so that the following case is avoided: a change in height of the lawn when the self-moving lawn mower moves causes an offset of the first positional image information **323** and the second positional image information **328** acquired by the image sensor **321**, and the acquired accurate position information has a relatively large error. The height sensor **326** may also be used for calibration of the angle of the self-moving lawn mower detected by the inertial sensor **325** with an excessively large offset due to the change in height.

In addition to the GPS positioning assembly or the GNSS positioning assembly mentioned above, the first positioning unit **310***a* may also be a displacement sensor **312** configured to detect the rotational speed or displacement of the moving wheel **130***a*. The displacement sensor **312** may be disposed on or near the drive motor **140***a* connected to the moving wheel **130***a*. The displacement sensor **312** acquires the rotational speed of the moving wheel **130***a* by detecting the rotational speed or displacement of the drive motor **140***a*. Alternatively, the displacement sensor **312** is disposed on or near the moving wheel **130***a* and configured to detect the rotational speed or displacement of the moving wheel **130***a* so as to acquire the main position estimation of the self-moving lawn mower.

For example, two moving wheels **130***a* are provided, which are a first moving wheel **131***a* and a second moving wheel **132***a*. Correspondingly, the drive motor **140***a* includes a first drive motor **141***a* and a second drive motor **142***a*. At least two displacement sensors **312** are provided and respectively disposed in the first moving wheel **131***a* and the second moving wheel **132***a* or respectively disposed in the first drive motor **141***a* and the second drive motor **142***a*, so as to acquire a rotational speed of the first moving wheel **131***a* and a rotational speed of the second moving wheel **132***a*. The rotational speeds and numbers of rotations of the first moving wheel **131***a* and the second moving wheel **132***a* are analyzed so that the main position estimation of the self-moving lawn mower is acquired. In conjunction with the moving trajectory obtained by the image sensor and the angular offset information of the moving trajectory of the self-moving lawn mower detected by the inertial sensor **325**, an accurate relative position of the self-moving lawn mower is acquired. Optionally, a speed difference between the first moving wheel **131***a* and the second moving wheel **132***a* is analyzed so that a displacement angle of the self-moving lawn mower can be acquired. In conjunction with the rotational speeds and the numbers of rotations of the moving wheels **130***a*, the accurate relative position of the self-moving lawn mower is acquired.

The control unit includes at least a main position estimation unit configured to establish a main position estimation function $g(x, y)$ of the self-moving lawn mower to obtain the main position estimation of the self-moving lawn mower. The control unit further includes an auxiliary position estimation unit configured to establish an auxiliary position estimation function h(x, y) according to the positional image information obtained by the image sensor to compensate for the main position estimation obtained by the main position estimation unit and obtain a final position estimation of the self-moving lawn mower, where the control unit is configured to drive at least one moving wheel to move toward a target position according to at least a target position instruction and the obtained final position estimation.

Optionally, the control unit **200***a* establishes the main position estimation function g(x, y) of the self-moving lawn mower according to the positioning data of the first positioning unit to obtain primary position data or the main position estimation of the self-moving lawn mower. For example, it is acquired that the self-moving lawn mower is in a certain region. The auxiliary position estimation function h(x, y) is established by using position information of a second detection unit. For example, a moving trajectory of the self-moving lawn mower in this region is acquired so as to compensate for the obtained primary position data or main position estimation data and obtain final accurate position data of the self-moving lawn mower. For example, the control unit establishes the main position estimation function g(x, y) of the self-moving lawn mower according to the displacement data detected by the displacement sensor and/or the positioning data of the positioning sensor and obtains the main position estimation of the self-moving lawn mower. The auxiliary position estimation function h(x, y) is established by using the position information obtained by the image sensor, the inertial sensor, and the height sensor. In conjunction with the main position estimation, the final position estimation of the self-moving lawn mower is obtained.

The image sensor **321** includes the film and the imaging device. The imaging device may be configured to be a camera, including a lens **324** and a package **325** for installing the lens. The film may be an imaging substrate or a photosensitive surface **327**. The imaging device senses an object to be measured projected by the self-moving lawn mower, and the corresponding first positional image information **323** and second positional image information **328** are generated on the film so that the displacement state of the self-moving lawn mower is analyzed by using the moving trajectory of one or more common feature area of the first positional image information **323** and the second positional image information **328**. The image sensor may further include a control chip or a PFE board and may process the generated first positional image information and second positional image information **328** to analyze and calculate the displacement information of the self-moving lawn mower or the object to be measured.

When the self-moving lawn mower moves forward and performs mowing, a casing **111***a* has a front end corresponding to an unmowed region of the mowing element **110***a* and a rear end corresponding to a mowed region of the mowing element **110***a*. The casing **111***a* or the main body has a first edge at the front end of the casing **111***a*, a second edge at the rear end of the casing **111***a*, and a third edge and a fourth edge between the front end and the rear end. The first edge, the second edge, the third edge, and the fourth edge form a boundary of the casing **111***a* of the self-moving lawn mower.

Optionally, the image sensor acquires the positional image information through the corresponding image sensing region, the image sensing region is disposed at a rear end of the main body, and the image sensor acquires the first positional image information and the second positional image information of the image sensing region in chronological order, where the first positional image information and the second positional image information have at least one common feature area. The control unit is configured to obtain, through the analysis, the relative displacement of the self-moving lawn mower according to at least the temporal moving trajectory of the at least one common feature area of the first positional image information and the second positional image information of the image sensor.

Optionally, referring to FIG. **20**, the image sensor **321** is disposed at the rear end **112***a* of the casing **111***a* or at a rear end of the self-moving lawn mower and connected to the information collection module **400***a*. The angle or steering of the image sensor **321** may be adjusted to be relatively downward so that the image sensing region **322** acquired by the self-moving lawn mower is the ground. When the self-moving lawn mower moves and mows the lawn, an image corresponding to the image sensing region **322** is the lawn and the first positional image information **323** and the second positional image information **328** are changing images of the lawn. The image sensor **321** is disposed at the rear end of the casing **111***a*, which corresponds to the mowed region of the self-moving lawn mower, and the grass is flattened by the self-moving lawn mower passing through the grass so that the lawn with different heights can be effectively prevented from affecting the acquisition of content of the image sensing region **322** by the image sensor **321**, thereby improving the accuracy of the displacement of the self-moving lawn mower detected by the image sensor **321**.

Referring to FIG. **22**, an image sensor **321***d* may be disposed near a rear side of moving wheels **130***d* or a rear side of grass pressing wheels of the self-moving lawn mower and is connected to the information collection module. The image sensing region generated by the image sensor **321***d* corresponds to a rear of the moving wheels **130***d* of the self-moving lawn mower, and the lens is disposed downward so that the ground or the vegetation can enter the image sensing region. When the self-moving lawn mower moves and performs mowing, the image sensor **321** acquires an image of the lawn that is pressed by the moving wheels **130***d* of the self-moving lawn mower so that the image sensing region acquired by the image sensor **321***d* may be less blocked by grass with different heights or relatively large heights, improving the accuracy of the first positional image information and the second positional image information acquired by the image sensor **321***d* and improving the accuracy of determination of the displacement of the self-moving lawn mower. Optionally, a distance between the image sensor **321***d* and the moving wheels **130***d* is greater than or equal to 1.5 cm and less than or equal to 3.5 cm. Optionally, the distance between the image sensor **321***d* and the moving wheels **130***d* is less than 2 cm. In this manner, the vegetation is pressed and lowered by the moving wheels **130***d* so that the interference of the vegetation on the detection of the image sensor **321***d* can be reduced. After the image sensor is disposed at this position, the accuracy of positioning the self-moving lawn mower by the image sensor can be effectively improved, and the error can be controlled to be within a proper accuracy range, for example, about 3%.

In an example, referring to FIG. **23**, the self-moving lawn mower is provided with a grass pressing piece **500**. Optionally, the grass pressing piece **500** is disposed on moving wheels **130***e*, for example, a grass pressing surface or a convex surface is formed on the moving wheels **130***e*.

Alternatively, the grass pressing piece **500** is connected to a casing **111**e of the self-moving lawn mower. The grass pressing piece **500** forms the grass pressing surface for pressing the lawn. The grass pressing surface has a preset width. In this manner, when the self-moving lawn mower moves on the vegetation, a compaction surface **510** is formed on the vegetation through the grass pressing surface of the grass pressing piece **500** so that a surface of the lawn is lowered. Moreover, an image sensing region of an image sensor **321**e corresponds to the compaction surface **510** generated by the current self-moving lawn mower, and at least part of the compaction surface is located in the image sensing region. The lowered lawn can reduce the interference of the height of the grass on the acquisition of the first positional image information and the second positional image information by the image sensor **321**e so that the acquired first positional image information and second positional image information are clear and complete, thereby improving the detection accuracy of the position of the self-moving lawn mower. Exemplarily, the moving wheels may be directly made into the grass pressing wheels, that is, the grass pressing piece is disposed on the moving wheels so that the interference of the height of the grass on the image sensor parsing image information is reduced. The grass pressing piece is connected to the rear end of the casing, and the image sensor is disposed at the rear end of the casing. In this manner, as the self-moving lawn mower moves, the grass pressing piece presses the vegetation, and the image sensor detects the pressed vegetation so that the interference of the vegetation with different heights on the acquisition of the positional image information by the image sensor is reduced, thereby improving the detection accuracy of the image sensor. Optionally, the height sensor is also disposed at the rear end of the casing and near the image sensor.

In an example, referring to FIG. **24**, the self-moving lawn mower forms an airflow on a main body **100**f through an airflow generation device **600** and makes the airflow flow through a surface of an image sensor **321**f. Since the image sensor **321**f determines the displacement state of the self-moving lawn mower by the displacement of the common feature area of the first positional image information and the second positional image information, relatively high requirements are put on the imaging quality of the first positional image information and the second positional image information. During operation, the self-moving lawn mower moves and performs mowing, and grass clippings and dust brought by the mowing are attached to the image sensor **321**f. Long-term accumulation of the grass clippings and dust will seriously affect the imaging effect of the image sensor **321**f, and the dust or grass clippings caused by the mowing operation of the self-moving lawn mower will block the image sensing region, affecting the imaging result and interfering with the determination of the displacement of the self-moving lawn mower. A distance between the airflow generation device and the image sensor is configured to be greater than or equal to 2 cm and less than or equal to 8 cm, and a flow direction of the airflow generated by the airflow generation device is away from the imaging device. The airflow generation device **600** may be a fan disposed near the image sensor **321**f and continuously generate the airflow when the self-moving lawn mower is in operation, so as to reduce the interference of the dust or grass clippings on the detection of the image sensor **321**f.

The main body **100**f includes an air inlet, an air outlet, and an airflow channel guiding the airflow from the air inlet to the air outlet. The image sensor **321** is disposed near the air outlet of the airflow channel, and the surface of the image sensor **321** is impacted by the airflow flowing out of the air outlet. The airflow in the airflow channel may be generated through the moving state of the self-moving lawn mower or may be generated through the operation of the airflow generation device **600**, such as the fan, disposed in the self-moving lawn mower or on a casing **111**f.

Referring to FIG. **25**, in an example, to optimize an image sensor **321**g, a light source supplementing device **700** and a light source filtering device **710** may be provided. The light source supplementing device **700** may emit light with a single spectrum, for example, a light source such as a laser, and project the light to an image sensing region **322**g. The light source filtering device **710** is disposed in the front of the image sensor **321**g and configured to filter other light sources and only allow the light with a single spectrum to pass through, for example, the light source such as the laser. The first positional image information and the second positional image information of detection light with a single spectrum emitted by the light source supplementing device **700** are acquired so that the displacement information of the self-moving lawn mower is acquired, thereby improving the detection accuracy of the displacement and position of the self-moving lawn mower. Optionally, a light source with a single spectrum of light is transmitted through the light source supplementing device **700** so as to emit light with a single spectrum to the vegetation through which the self-moving lawn mower passes; the light source filtering device **710** is configured to allow a light source with the same spectrum as a detection light source transmitted by the light source supplementing device **700** to pass through so that only the light with the same spectrum can be acquired, detected, and analyzed by the image sensor **321**g, thereby effectively reducing the influence of outdoor ambient light on a measurement accuracy of an optical flow sensor.

Optionally, the self-moving lawn mower further includes an ultrasonic sensor and a collision sensor configured to detect an obstacle. Both the ultrasonic sensor and the collision sensor are communicatively connected to the information collection module **400**. The ultrasonic sensor transmits an ultrasonic wave, detects whether an obstacle exists on a preset route of the self-moving lawn mower, and records position information of the obstacle. The ultrasonic wave bounced by the obstacle is sensed by the ultrasonic sensor, and the position of the obstacle is analyzed and acquired by a time interval. When the self-moving lawn mower hits the obstacle, the collision sensor feels shaking of the casing **111**a or a pressure change of the casing **111**a and analyzes whether the self-moving lawn mower hits the obstacle. Therefore, the collision sensor may be configured to be a Hall sensor configured to detect a displacement state or a pressure sensor configured to detect a pressure change.

Optionally, the final accurate position information of the self-moving lawn mower is acquired by using only one or more image sensors, that is, the combination of the first positioning unit and the second positioning unit is no longer used for obtaining the final accurate position information, and only a combination of one or more image sensors is used for directly acquiring the final accurate position information. One or more image sensors may be used for acquiring a 3G three-dimensional scene at the position of the self-moving lawn mower, thereby acquiring positioning information of the self-moving lawn mower. For example, one or more image sensors may collect the 3G three-dimensional scene at the position of the self-moving lawn mower and interact with map interfaces such as AutoNavi Map or Google Map. In this manner, actual positioning information of the self-moving lawn mower is obtained, and then the control unit of the self-moving lawn mower drives at least one moving wheel to move toward a target position according to at least a target position instruction and the obtained final position estimation. The map interfaces such as AutoNavi Map or Google Map may be directly disposed on a display screen of the self-moving lawn mower or on a display screen of a mobile device such as a mobile phone. Exemplarily, one or more image sensors may be movably steered to obtain a 3G three-dimensional scene or a 360° three-dimensional scene of a predetermined region of the self-moving lawn mower, which is not limited here.

Referring to FIG. **26**, the self-moving lawn mower further includes multiple image sensors **321***h* disposed at at least two of a front end **113**, a rear end, or a side end **114** of the main body.

The image sensors **321***h* are disposed near a peripheral edge of the main body and configured to sense first positional image information and second positional image information formed by an object to be measured in a preset reference plane in chronological order, where the first positional image information and the second positional image information have at least one common feature area, the at least one common feature area includes one or more feature points, and the peripheral edge of the main body includes a first edge, a second edge, a third edge, and/or a fourth edge. One or more image sensors may be disposed on a top side of the main body or a main casing, and position information of the self-moving lawn mower or the object to be measured may also be obtained. The object to be measured is the vegetation or the obstacle to be measured on the ground. Optionally, the first edge, the second edge, the third edge, and the fourth edge not only refer to the boundary of the self-moving lawn mower but may also include a certain position near the boundary of the lawn mower and at a certain distance from the boundary of the intelligent lawn mower. Optionally, the image sensor disposed on the top side of the main body or the main casing can acquire the position information or positioning information of the self-moving lawn mower or the object to be measured by acquiring the 3G three-dimensional scene at the position of the self-moving lawn mower or the 3G three-dimensional scene of the object to be measured.

Information about the obstacle may be acquired through the image sensors **321***h*. One or more image sensors may be disposed on the peripheral edge of the main body of the self-moving lawn mower, such as the first edge, the second edge, the third edge, and/or the fourth edge. One or more image sensors acquire the information of the image sensing region and analyze whether the obstacle exists in the image sensing region or a distance between the obstacle and the self-moving lawn mower, that is, position information of the obstacle. Optionally, at least two image sensors may be provided, or an image sensing device is provided with a binocular camera, and the position information of the obstacle is acquired through the fusion of different positional image information collected by the binocular camera. Based on the preceding examples, the control unit analyzes the route of the self-moving lawn mower according to the analyzed position information of the self-moving lawn mower, a control instruction, and the tracked displacement state of the self-moving lawn mower, drives at least one moving wheel to move toward the target position, and drives a mowing blade to rotate.

To sum up, a positioning process for detecting a position of a self-moving lawn mower is provided. Referring to FIG. **29**, in step S**51**, a positioning sensor detects a position of the self-moving lawn mower so that a main position estimation of the self-moving lawn mower is acquired; in step S**52**, an inertial sensor detects a posture of the self-moving lawn mower; in step S**53**, an image sensor senses a region through which the self-moving lawn mower passes, forms some pieces of positional image information, and senses and forms first positional image information and second positional image information in chronological order; in step S**54**, at least one common feature area of the first positional image information and the second positional image information is acquired; in step S**55**, a height sensor detects a height parameter of the self-moving lawn mower relative to the positional image information; in step S**56**, a displacement of the self-moving lawn mower is calculated in conjunction with a temporal moving trajectory of the common feature area, the obtained posture of the self-moving lawn mower, and the height parameter; in step S**57**, a final position estimation of the self-moving lawn mower is calculated in conjunction with the displacement of the self-moving lawn mower acquired by the image sensor; in step S**58**, whether the self-moving lawn mower continues to move is detected. If so, step S**51** is repeated. If not, the process ends.

What is claimed is:

1. A self-moving lawn mower, comprising:

a main body comprising a casing;

a mowing element connected to the main body and configured to cut vegetation;

an output motor configured to drive the mowing element;

moving wheels connected to the main body;

a drive motor configured to drive the moving wheels to rotate; and

a control unit connected to the output motor and the drive motor and configured to control the output motor and the drive motor,

wherein the control unit is configured to:

identify all unmowed regions within an operation boundary;

generate an operation route for mowing in at least one unmowed region among the all unmowed regions; and

control the drive motor so that the self-moving lawn mower mows in the at least one unmowed region among the all unmowed regions according to the operation route, and

wherein the control unit is configured to identify the all unmowed regions within the operation boundary according to at least one of a moving trajectory of the self-moving lawn mower or a positional image information related to the moving trajectory.

2. The self-moving lawn mower of claim **1**, wherein a shortest route along which mowing is sequentially performed in the at least one unmowed region among the all unmowed regions is defined as a shortest operation route and a ratio of a length of the operation route to a length of the shortest operation route is greater than or equal to 1 and less than or equal to 1.2.

3. The self-moving lawn mower of claim **1**, wherein the control unit comprises a filling planning module configured to calculate a shortest operation route for the self-moving lawn mower to supplementary operate the all unmowed regions.

4. The self-moving lawn mower of claim **3**, wherein the control unit is configured to control the self-moving lawn mower to sequentially perform the supplementary operation on the all unmowed regions along the shortest operation route.

5. The self-moving lawn mower of claim **4**, wherein after the self-moving lawn mower performs the supplementary operation on each of the all unmowed regions and a coverage rate of the supplementary operation the self-moving lawn mower is controlled to perform on the each of the all unmowed regions is greater than 80%.

6. The self-moving lawn mower of claim **1**, wherein the self-moving lawn mower further comprises a positioning assembly and the positioning assembly comprises one of or a combination of a global positioning system (GPS) positioning unit, an inertial measurement unit (IMU), a displacement sensor, and an image sensor.

7. The self-moving lawn mower of claim **6**, wherein the positioning assembly acquires an operation trajectory of the self-moving lawn mower, a non-operated region of the self-moving lawn mower within the operation boundary is determined according to the operation trajectory of the self-moving lawn mower and information about the operation boundary and, in a case where an area of the non-operated region is greater than a preset value, the non-operated region is determined to be an unmowed region.

8. The self-moving lawn mower of claim **1**, wherein the self-moving lawn mower further comprises an image sensor and the image sensor is configured to acquire a two-dimensional image or a three-dimensional image within the operation boundary to acquire information about the all unmowed regions operated by the self-moving lawn mower.

9. The self-moving lawn mower of claim **1**, wherein the self-moving lawn mower further comprises a positioning assembly comprising at least one of an image sensor configured to sense the moving trajectory or the positional image information of the self-moving lawn mower.

10. The self-moving lawn mower of claim **9**, wherein the image sensor is configured to sense and form a first positional image information and a second positional image information in chronological order, the first positional image information and the second positional image information have at least one common feature area, and the control unit is configured to obtain a relative displacement of the self-moving lawn mower by analyzing at least a temporal moving trajectory of the at least one common feature area of the first positional image information and the second positional image information.

11. The self-moving lawn mower of claim **9**, wherein the image sensor is disposed behind the moving wheels.

12. The self-moving lawn mower of claim **9**, wherein a distance between the image sensor and one of the moving wheels is greater than or equal to 1.5 cm and less than or equal to 3.5 cm.

13. The self-moving lawn mower of claim **9**, wherein the image sensor comprises a lens and a package for installing the lens and the lens is arranged downward so that a ground or the vegetation is capable of entering an image sensing region.

14. The self-moving lawn mower of claim **1**, wherein the self-moving lawn mower further comprises an interactive interface communicatively connected to the control unit and operated so as to add or remove an unmowed region.

15. A supplementary operation method for unmowed region of a self-moving mowing system, comprising:

- selecting an operation region of the self-moving mowing system;
- starting the self-moving mowing system, positioning a self-moving lawn mower of the self-moving mowing system to acquire an operation moving trajectory, and determining that a coverage region of moving operation of the self-moving lawn mower is an operated region;
- acquiring a non-operated region in the operation region by analyzing the operation region and the operated region and determining that a non-operated region with an area greater than a preset value is the unmowed region;
- planning an operation route for supplementary operation on unmowed regions according to information about the unmowed regions; and
- controlling the self-moving lawn mower to perform the supplementary operation on at least one unmowed region among all the unmowed regions according to the operation route,
- wherein positioning the self-moving lawn mower of the self-moving mowing system comprises:
  - detecting, by an inertial sensor, a posture of the self-moving lawn mower;
  - sensing, by an image sensor, a region through which the self-moving lawn mower passes, forming positional image information, and sensing and forming a first positional image information and a second positional image information in chronological order;
  - acquiring at least one common feature area of the first positional image information and the second positional image information;
  - detecting, by a height sensor, a height parameter of the self-moving lawn mower relative to the positional image information, and
  - calculating a displacement of the self-moving lawn mower according to a temporal moving trajectory of the at least one common feature area, the posture of the self-moving lawn mower, and the height parameter.

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